

Unit 2

The Network Infrastructure for E-Commerce



- Information Superhighway
- Components of the I-Way
- Internet as network infrastructure



Introduction to Information Superhighway

Information Superhighway

The **Information Superhighway** is a term used to describe the internet and other digital communication networks that allow people to quickly access and share information. Just like highways help people travel from one place to another, the information superhighway helps data move from one computer to another, connecting people, businesses, and governments worldwide.



Information Superhighway

Global network of computers that potentially will connect most of world's individuals, firms, and organizations. It is envisioned to provide very **high speed** wireless connection.

Information Superhighway

Understand by Analogy

Imagine a big road (highway) where vehicles (cars, buses, trucks) travel from one place to another.

Similarly, the Information Superhighway is a digital road where information (messages, videos, files, websites) moves between computers, smartphones, and servers.

Just like highways connect cities and towns, the Information Superhighway (internet) connects people across the world.



Information Superhighway (Contd..)

The information superhighway is very much a physical network, an infrastructure of modern High-capacity, interactive electronic pipeline providing integrated services. I-way links everyone at home or office to everything else.

The information superhighway is a physical network, facilitating the broadband, two way transmission of any type of digital information, within its own virtual space.

Note

Any e-commerce application will need the I-way infrastructure in the same manner as any normal business would require the interstate highway network to carry goods from one place to other

Roles of I - Way in e-commerce

Facilitates global connectivity, enabling businesses to reach international markets.

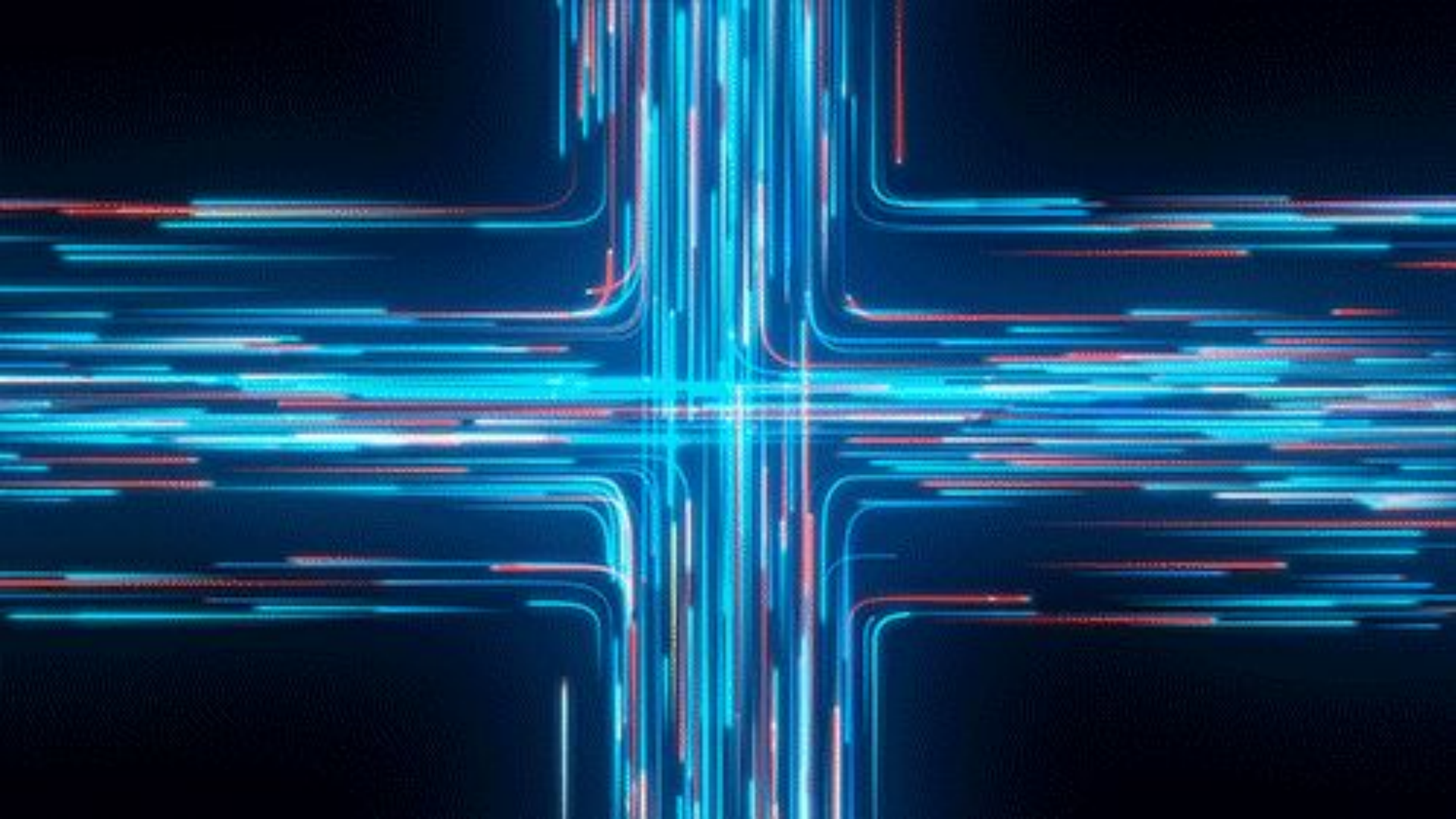
Supports seamless communication between buyers, sellers, and service providers.

Enables real-time transactions and instant payments through secure networks.

Provides a platform for online advertising and digital marketing.

Enhances customer service through chatbots, live support, and personalized recommendations.







US Net is building a national backbone. When completed in 2002, the network will interconnect its major metropolitan service areas with a fault tolerant diversely routed 155 Mbps clear channel fiber backbone. All Internet connectivity will be aggregated at MAE East, MAE West, ChiNap, and MAE Central, where it can be directly handed off to US Net's major peering partners





Nepal Telecom

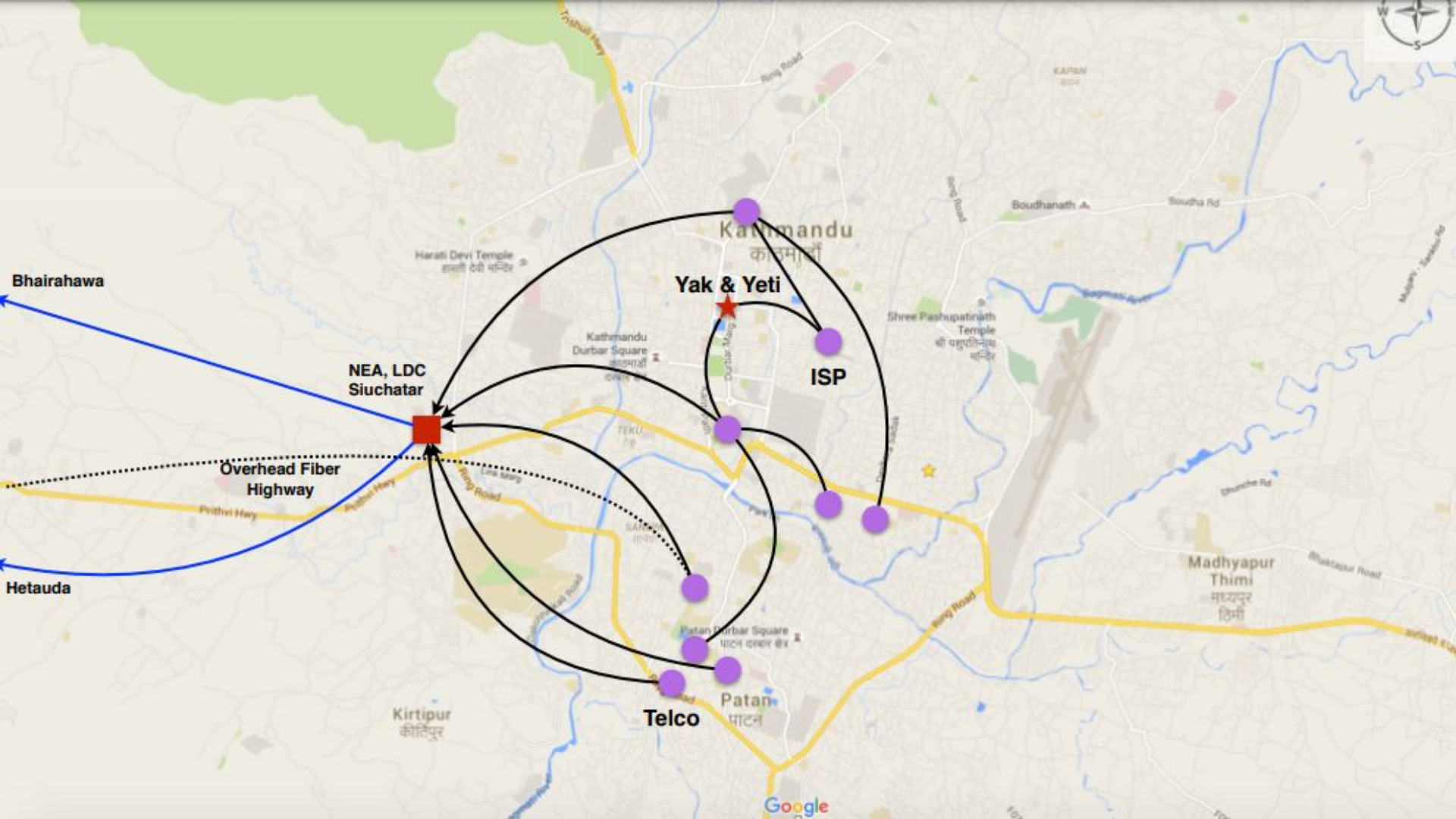
Transmission Planning Department

OPTICAL FIBER BACKBONE NETWORK OF NEPAL TELECOM



Prepared on : 14 December, 2006

By: Ananta Mallik







China
Tibet

Shey
Phoksundo
National Park
शेय
फोक्सुण्डो
राष्ट्रिय
पार्क

Annapurna
Conservation
Area
अन्नपूर्ण
संरक्षण
क्षेत्र

Nepal

Rasuwagadi

Tatopani

Bhairawa

Birgunj

Biratnagar

India

SIKKIM

Gangtok

Darjeeling

Siliguri

Kathmandu
काठमाडौं

Lakhimpur

Sitapur

Barabanki

Lucknow

Faizabad

Basti

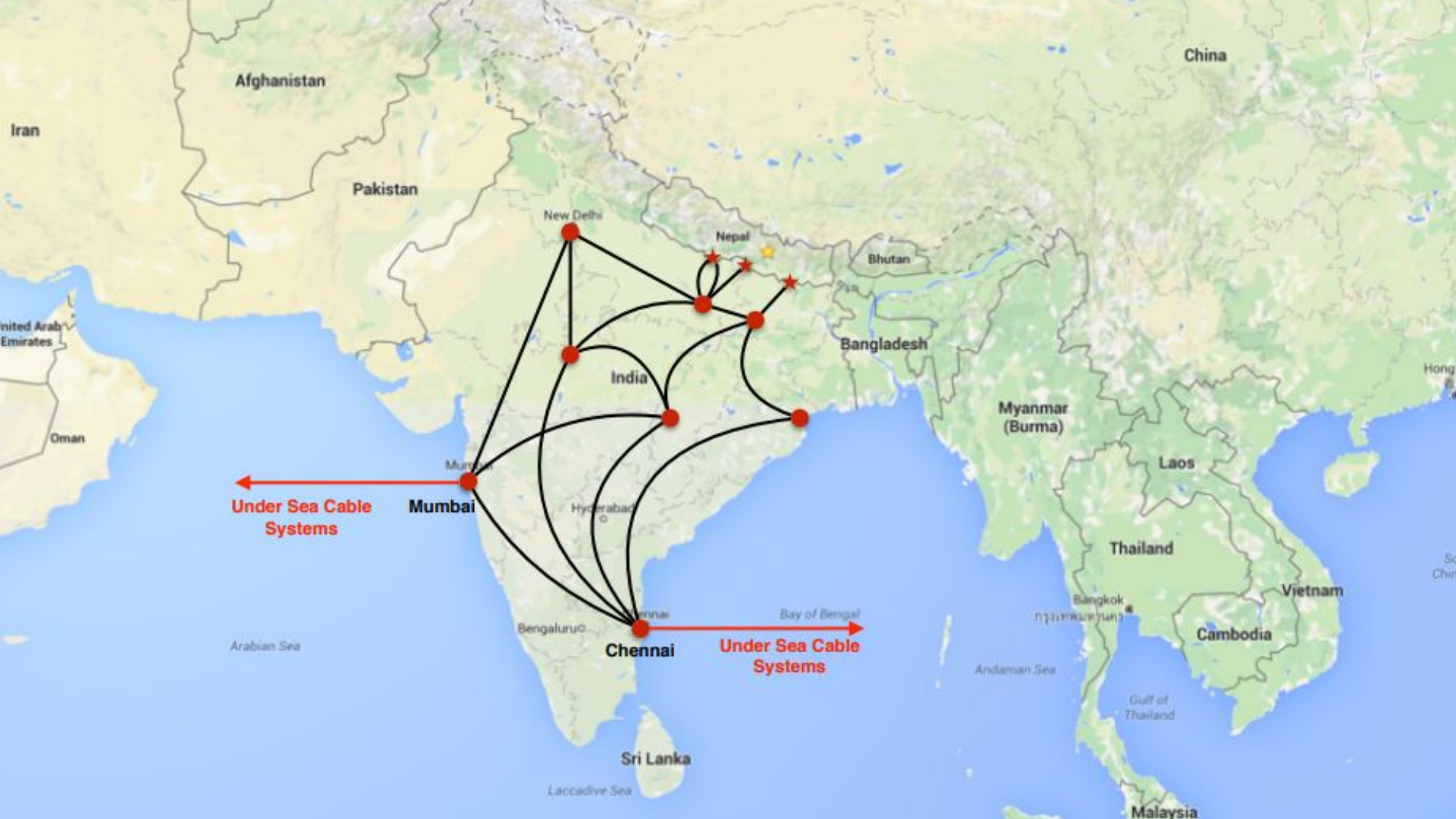
Kushinagar

Kuchakote

Deoria

Madhubani

Kanpur

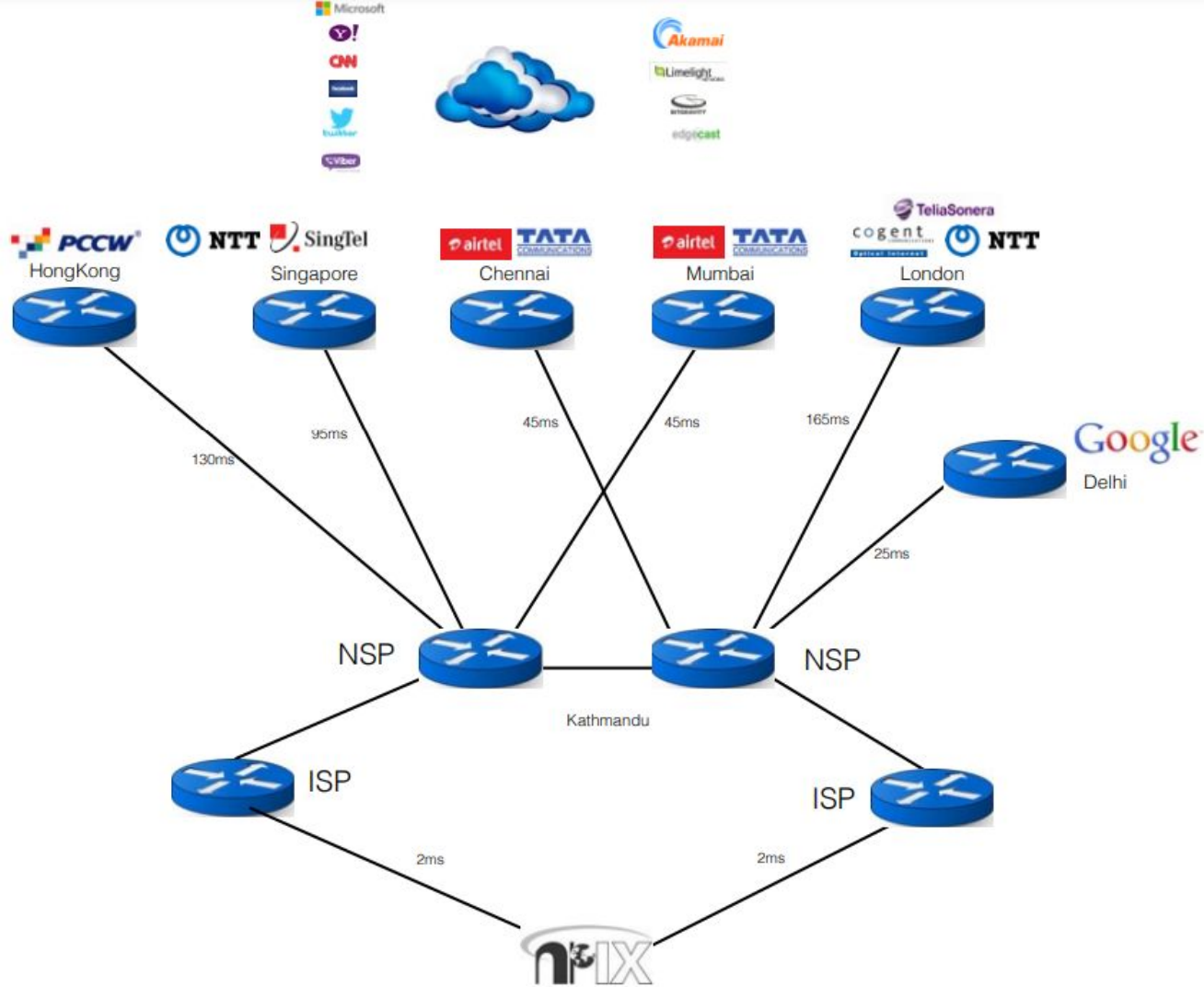


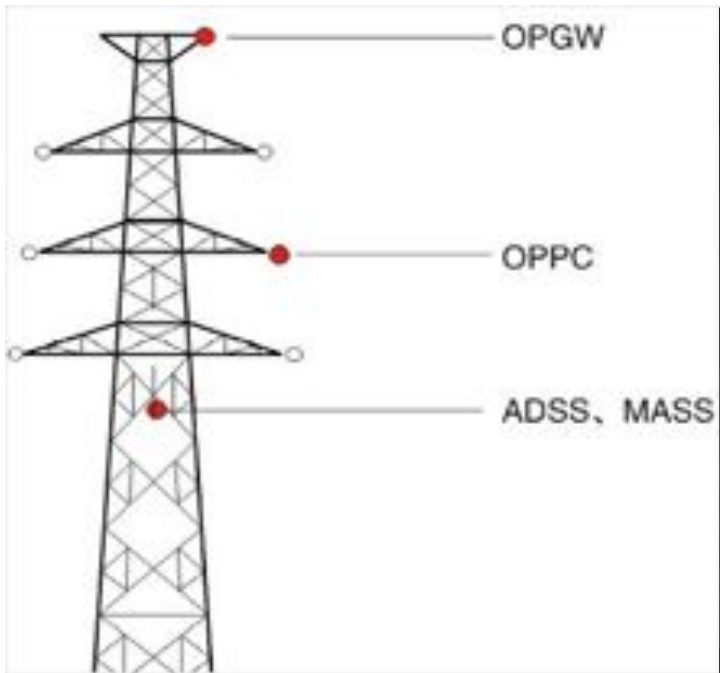




NPIX is the internet exchange Nepal where all the ISP, telecom operator and content delivery network (CDN) gets connected and exchange the traffic between them. NPIX facilitates keeping local traffic local, making local surfing much more desirable with local content for local customers, at the same time saving international bandwidth.

NPIX is the nepal's own internet exchange point which is located in two places, Jawalakhel and Putalisadak. Every country have their own internet exchange point for providing the internet exchange services.





A woman with short dark hair, wearing a dark blue polo shirt, is seated at a desk and looking at a vintage computer monitor. She is smiling slightly. The monitor displays a text-based interface with several lines of text. On the desk, there is a keyboard and a mouse. In the background, a window shows a view of a residential building with multiple windows and some greenery. The word "THAMES" is overlaid in large white letters in the bottom left corner.

THAMES

When I fix the WiFi

What my parents
think I do



What I really
do



**my parents who don't
know how to connect
wifi**

me:

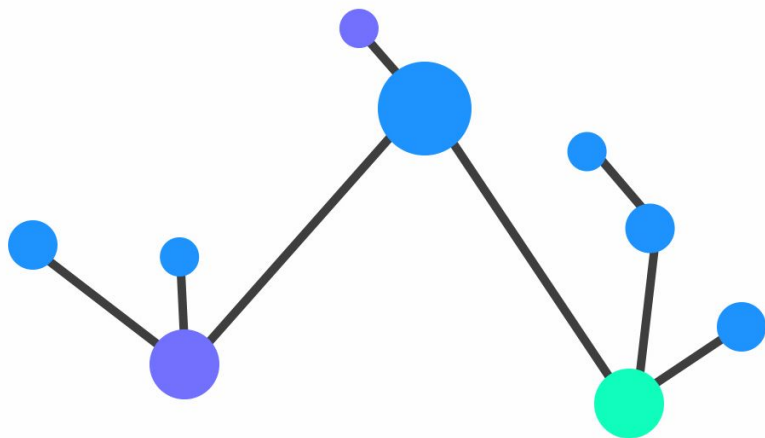


An extensive electronic network such as the internet, used for the rapid transfer of information such as sound, video, and graphics.





Components of Information Superhighway



iWay components

Consumer access equipment

Local or access road, or
on-ramps

Global information
Distribution Network

**Consumer
Access Equipment**

Local on-Ramps

**Global Information
Distribution Networks**

**Consumer /
Business
premises
equipment**

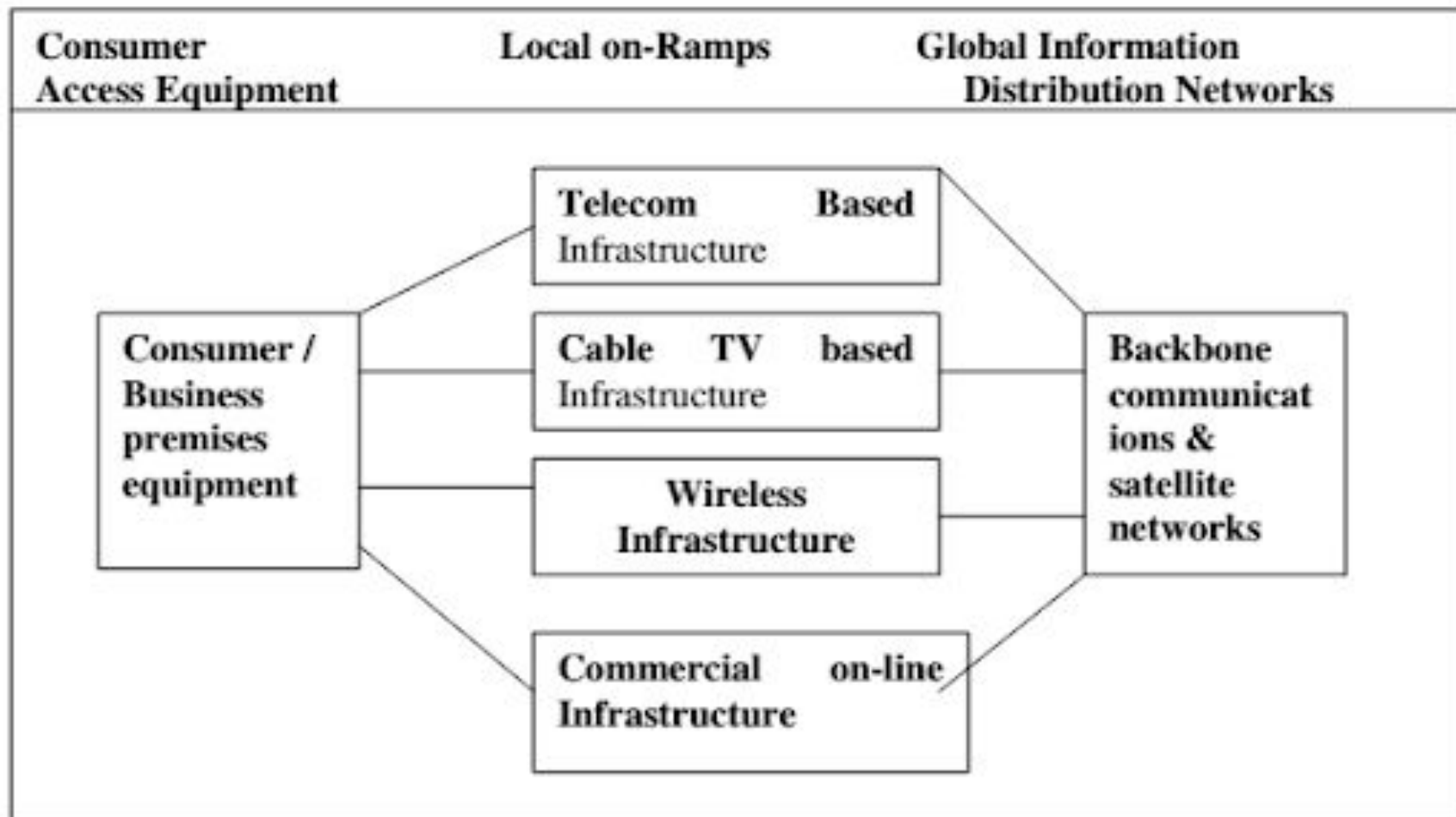
**Telecom Based
Infrastructure**

**Cable TV based
Infrastructure**

**Wireless
Infrastructure**

**Commercial on-line
Infrastructure**

**Backbone
communicat
ions &
satellite
networks**



Customer-Access Equipment

— — —

Hardware such as routers, network switches, gateways, set-top boxes, home networking adapters.

Software such as browsers and OS.



Consumer Access Equipment

These are the devices that individuals use to connect to the internet and digital services, such as smartphones, computers, tablets, routers, and smart TVs.

It includes **devices** such as: Smartphones, Laptops and desktops, Tablets, Smart TVs, Routers and modems (which connect devices to the internet)

Consumer Access Equipment

These are the tools we use to connect to the internet and access digital information.

Examples: Smartphones, laptops, desktop computers, tablets, smart TVs, and gaming consoles.

How it works: When you use a phone to search for something online, your device sends a request through the internet to a server, which then sends back the information.

Local-on ramps

Wires linking homes with the backbone network.

Requires huge investment.

Also known as “last mile”



Local-on ramps

These are the network connections that link users to the internet, including Wi-Fi, mobile data, broadband services, and local area networks (LANs).

Examples include: Wi-Fi connections at home or in public places, Mobile data networks (3G, 4G, 5G), Broadband services (fiber optics, DSL, cable internet), Local area networks (LANs) in offices and schools

Global Information Distribution Network

This is the worldwide infrastructure that enables data transfer and communication, consisting of fiber optic cables, satellites, undersea cables, and data centers that store and distribute information globally.

It **includes**: Fiber optic cables that transmit data between countries and continents, Satellites that help with global communication and internet access, Data centers and cloud servers that store and distribute information, Undersea cables that connect different parts of the world.

In short ..

— — —

Consumer Access Equipment = Devices like phones or computers

Local or Access Road = Internet connections (Wi-Fi, broadband, etc.)

Global Information Network = The massive worldwide internet system



Internet as network infrastructure

Write about Internet as Network Infrastructure in your own
word - Assignment

Network Infrastructure

Network infrastructure refers to the hardware, software, and technologies that enable communication between devices and the internet. It forms the backbone of the internet and private networks, allowing data to move efficiently.

Let's understand NIF first..

— — —

Network infrastructure refers to the physical and virtual components that enable communication and data transfer between devices, systems, and networks. It includes hardware, software, and services that support connectivity, management, and security.

Key Components of Network Infrastructure

— — —

Hardware:

- Routers
- Switches
- Modems
- Servers
- Cables (Ethernet, fiber optics, etc.)
- Network Interface Cards (NICs)

Software:

- Network operating systems
- Firewalls
- Security protocols
- Network management tools

Key Components of Network Infrastructure (Contd..)

Services & Protocols:

- Internet Service Providers (ISPs)
- Cloud services
- DNS (Domain Name System)
- VPN (Virtual Private Network)
- TCP/IP, HTTP, FTP, etc.

Explanation..

Routers – Think of a router as a traffic controller for the internet. It directs data between different networks, such as your home Wi-Fi and the internet.

Switches – It connects multiple devices within a network (like computers, printers, and servers) so they can share information.

Explanation..

Modems - The modem is your gateway to the internet. It converts signals from your Internet Service Provider (ISP) into a format your network can use.

Servers - These are powerful computers that store data, applications, and websites, making them accessible to users in a network.

Cables (Ethernet/Fiber Optic) - Just like roads for cars, cables allow data to travel between devices in a wired network.

Network Interface Cards (NICs) - A small chip inside a computer that allows it to connect to a network, either through cables or wirelessly.

Explanation..

Network Operating Systems – Special software (like Windows Server or Linux) that manages network devices and controls data flow.

Firewalls – A security system that blocks unauthorized access to your network, like a gatekeeper protecting your house.

Network Management Tools – Software that helps monitor and maintain network performance, preventing slow speeds and connectivity issues.

Explanation..

DNS (Domain Name System) – Converts website names (like google.com) into numerical IP addresses, so computers can understand where to send requests.

VPN (Virtual Private Network) – Creates a secure, encrypted connection over the internet, protecting your data from hackers.

TCP/IP (Transmission Control Protocol/Internet Protocol) – A set of rules that ensures data is sent and received correctly over the internet.

What are parts of network infrastructure that help in data transmission and internet connectivity?



Term	What it does	Example
Backbone	Main high-speed connection	Global Internet Cables
Bandwidth	Data Transfer Capacity	100 Mbps vs. 2 Mbps speed
NAP	Connects Networks	Early internet NAPs
MAE	City wide network Exchange	MAE-East in Washington D.C.
Campus Network	Private Network for an institution	Harvard's Internal network
ISP	Provides internet	Comcast, AT&T, Verizon
Narrowband	Slow internet	Dial-up internet
Broadband	Fast Internet	100 Mbps broadband
Cable modem	Connects cable internet	Xfinity cable modem
DSL	Internet via phone lines	AT&T DSL service

Backbone

High-bandwidth fiber optic cable that transports data across the Internet.

Network Service Provider (NSP) owns and controls one of the major networks comprising the Internet backbone.

Backbone

— — —

A backbone is the main high-speed connection in a network that carries large amounts of data between different smaller networks. It is responsible for keeping networks connected and ensuring data moves efficiently.

Analogy:

Think of the backbone as a high-speed expressway that connects different cities. While smaller roads lead to neighborhoods, the expressway carries most of the heavy traffic between major locations.

Example:

The internet backbone consists of high-capacity fiber optic cables that connect different countries and continents, allowing global communication.

Bandwidth

— — —

Measures how much data can be transferred over a communications medium within a fixed period of time; is usually expressed in bits per second (bps), kilobits per second (Kbps), or megabits per second (Mbps)

Bandwidth

Bandwidth refers to the amount of data that can be transferred over a network in a specific time. Higher bandwidth means faster internet speeds, while lower bandwidth results in slow connections.

Analogy:

Imagine bandwidth as a water pipe—a bigger pipe allows more water to flow at once, just like higher bandwidth allows more data to transfer at a faster rate.

Example:

If you have 100 Mbps bandwidth, you can stream multiple HD videos smoothly. If the bandwidth is low (e.g., 2 Mbps), streaming will buffer frequently.

Network Access Points and Metropolitan Area Exchanges

One of the hubs where the backbone intersects with regional and local networks, and where the backbone owners connect with one another

Metropolitan Area Exchanges (MAEs) another name for one of the hubs where the backbone intersects with regional and local networks.

Network Access Points

A Network Access Point (NAP) is a physical location where different internet networks connect and exchange data. These points help in efficient data routing across the internet.

Analogy:

Think of a NAP like an airport hub where flights from different cities meet, allowing passengers (data) to switch planes and reach their destinations.

Example:

In the early internet days, major NAPs were built in cities like New York and San Francisco, allowing ISPs to connect their networks.

Metropolitan Area Exchange

A Metropolitan Area Exchange (MAE) is a specific type of NAP that allows different ISPs and networks to exchange data within a city or metropolitan area.

Analogy:

It's like a central bus station where people from different routes transfer to different buses to reach different parts of the city.

Example:

MAE-East, located in Washington D.C., was one of the first large-scale internet exchange points that helped ISPs connect within the U.S.

Campus Networks

Generally local area networks operating with a single organization that leases access to the Web directly from Internet Service Providers

Campus Network

A Campus Network is a private network that connects multiple buildings within a campus, such as universities, corporate offices, or hospitals. It provides high-speed connectivity for internal communications.

Analogy:

A campus network is like an internal road system inside a large university. It connects all buildings efficiently without needing to go onto the public highway (internet).

Example:

Harvard University has its own campus network that connects classrooms, libraries, and research centers for fast communication and file sharing.

Internet Service Providers

Firm that provides the lowest level of service in the multi-tiered Internet architecture by leasing Internet access to home owners, small businesses, and some large institutions

ISP

— — —

An Internet Service Provider (ISP) is a company that provides internet access to homes and businesses. ISPs manage the infrastructure that connects customers to the global internet.

Analogy:

An ISP is like a water company that supplies water (internet) to your home. The company owns the pipelines (network infrastructure) and delivers the service.

Example:

Companies like Comcast, AT&T, and Verizon are major ISPs that provide internet access in different regions.

Narrowband

— — —

The traditional telephone modem connections, now operating at 56.6 Kbps

Narrowband

— — —

Narrowband refers to low-speed internet connections, typically less than 56 Kbps. It was used in early internet days and is much slower than modern broadband.

Analogy:

Narrowband is like a narrow hiking trail—only one or two people can walk at a time, so movement is slow and limited.

Example:

Dial-up internet, which used telephone lines and had a speed of 56 Kbps, is an example of narrowband internet.

Broadband

Refers to any communication technology that permits clients to play streaming audio and video files at acceptable speeds -- generally above 100 Kbps Broadband service is based on DSL, cable modem, telephone (T1 and T3 lines), and satellite technologies

Broadband

— — —

Definition:

Broadband is a high-speed internet connection that allows large amounts of data to be transmitted quickly. It includes cable, fiber optic, and satellite connections.

Analogy:

Broadband is like a multi-lane highway, where multiple cars (data packets) can move fast at the same time, reducing congestion.

Example:

If you have a 100 Mbps broadband connection, you can stream Netflix, play online games, and download files at the same time without lag.

Cable modem

It refers to a cable television technology that piggybacks digital access to the Internet using the same analog or digital video cable providing television signals to a home. Cable Internet is a major broadband alternative to DSL service, generally providing faster speeds and a "triple play" subscription: telephone, television, and Internet for a single monthly payment

Cable modem

A cable modem is a device that connects your home to the internet using cable TV lines. It converts signals from the ISP into a format that your devices can understand.

Analogy:

A cable modem is like a translator that converts one language (ISP signals) into another language (your computer's data format).

Example:

If you use Xfinity or Spectrum cable internet, you likely have a cable modem in your home that provides connectivity.

Digital Subscriber Line (DSL)

It is a telephone technology that provides high-speed access to the Internet through ordinary telephone lines. Found in a home or business. Service levels range from about 768 Kbps up to 7 Mbps

Digital Subscriber Line

— — —

Definition:

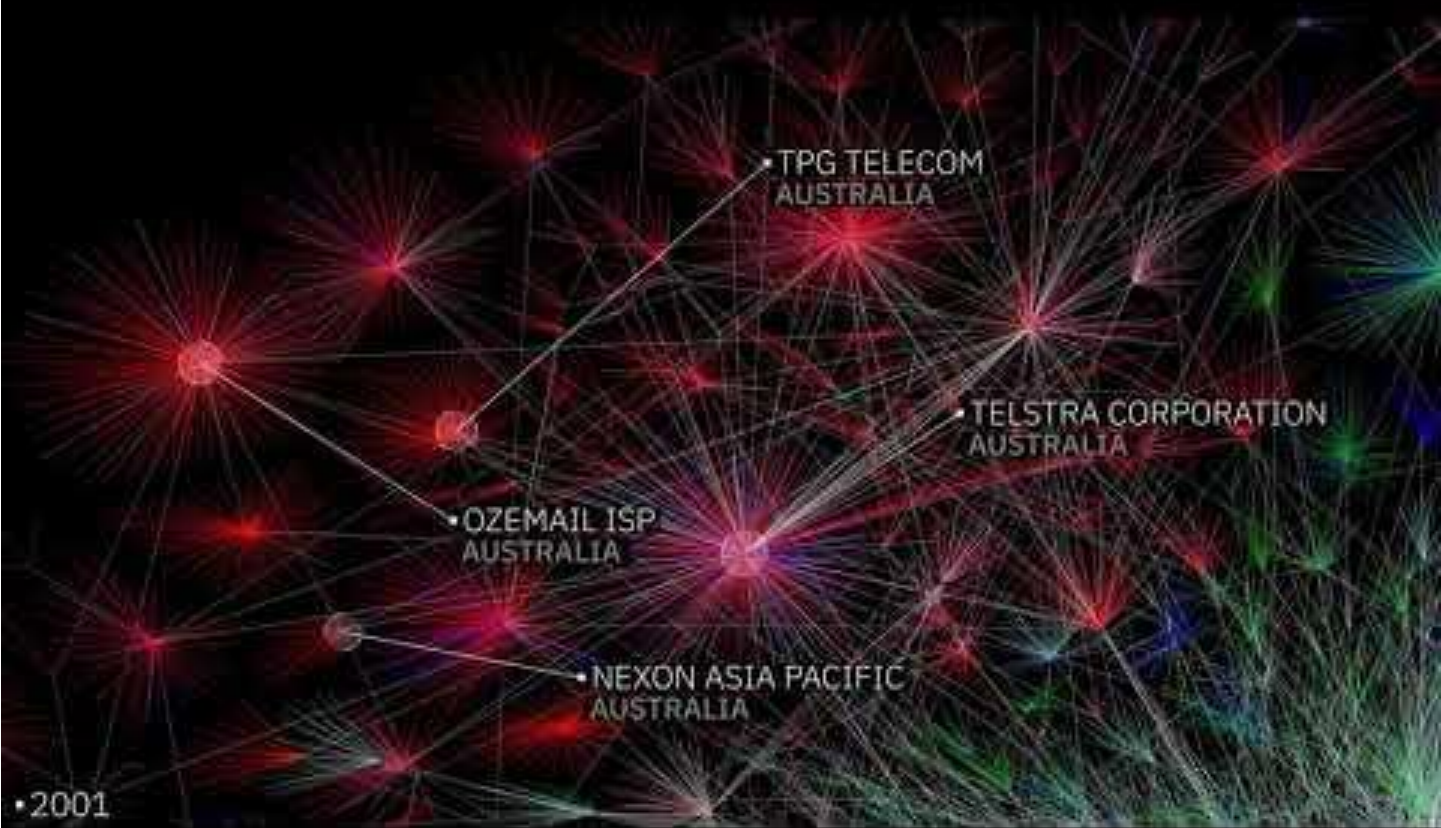
A Digital Subscriber Line (DSL) is an internet connection that uses telephone lines but is much faster than old dial-up internet.

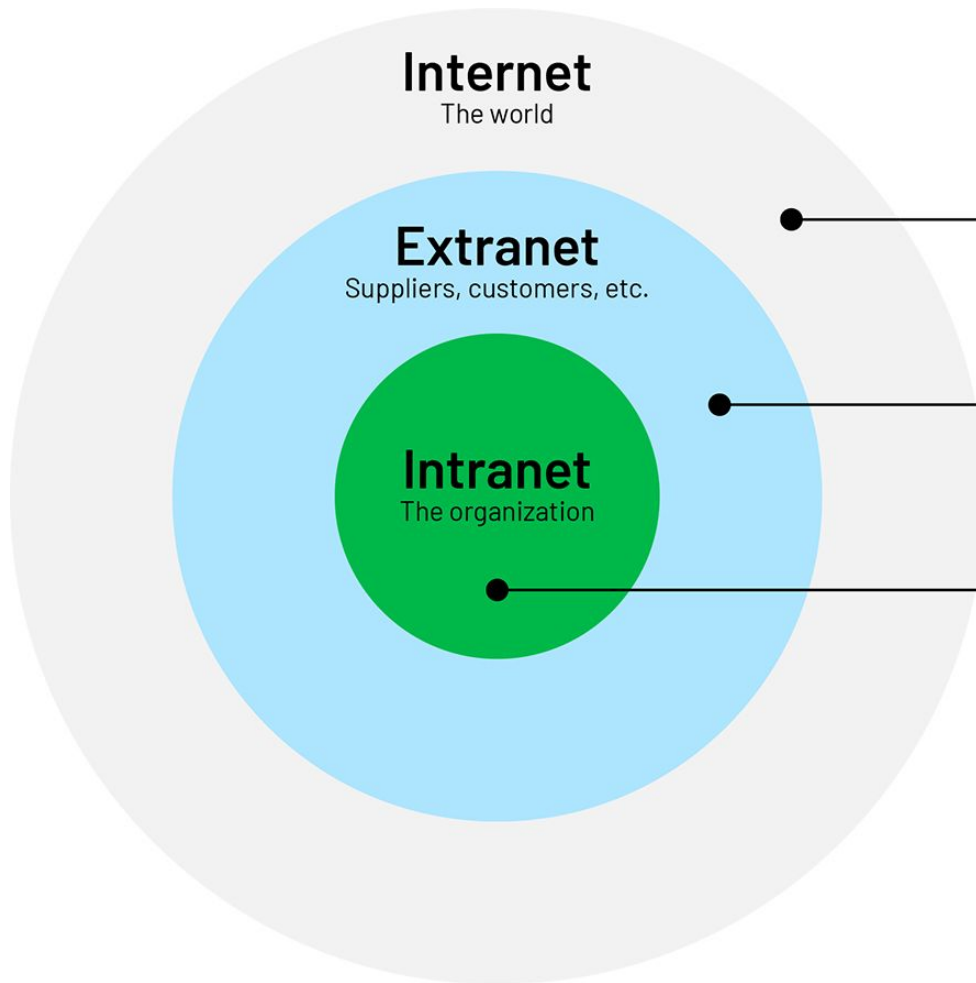
Analogy:

DSL is like a private lane on a road—it still uses the same infrastructure (telephone lines) but moves faster than the main traffic.

Example:

If you use AT&T's DSL internet, your internet connection runs through your phone line but does not interfere with phone calls.





Internet

The world

Extranet

Suppliers, customers, etc.

Intranet

The organization

The **internet** creates connections between computers around the world.

An **extranet** creates connections beyond (or outside) an organization.

An **intranet** creates connections inside an organization.

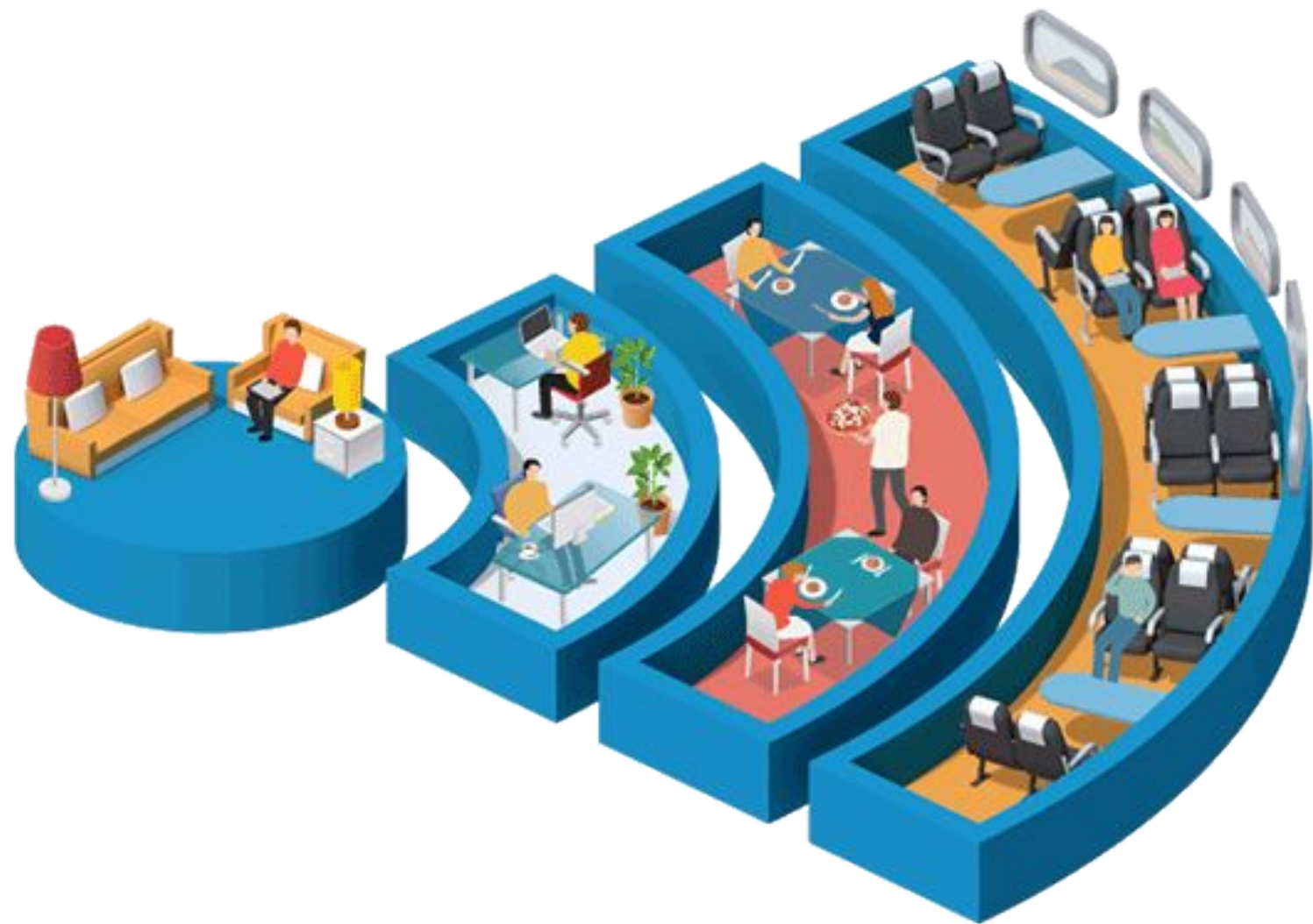
Unit 2

Wireless Application Protocol



Wireless Application Protocol

- Wireless Application Protocol (WAP)
- Architecture of WAP
- Working of WAP



WHEN YOUR WIFI



STOPS WORKING



WAP is a technical standard for transferring information to wireless devices.

Wireless Application Protocol

WAP Stands for Wireless Application Protocol

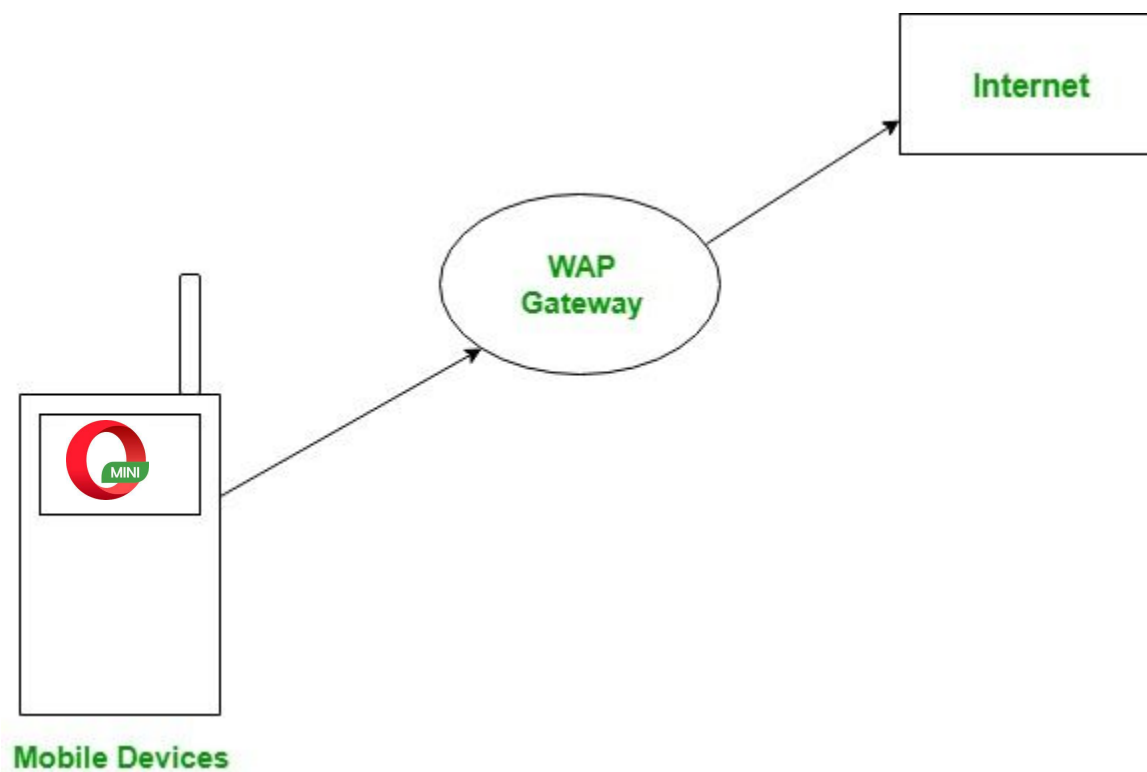
It is a protocol designed for micro-browsers and it enables the access of internet in the mobile devices.

It uses markup language WML (Wireless Markup Language and not HTML), WML is defined as XML 1.0 application.

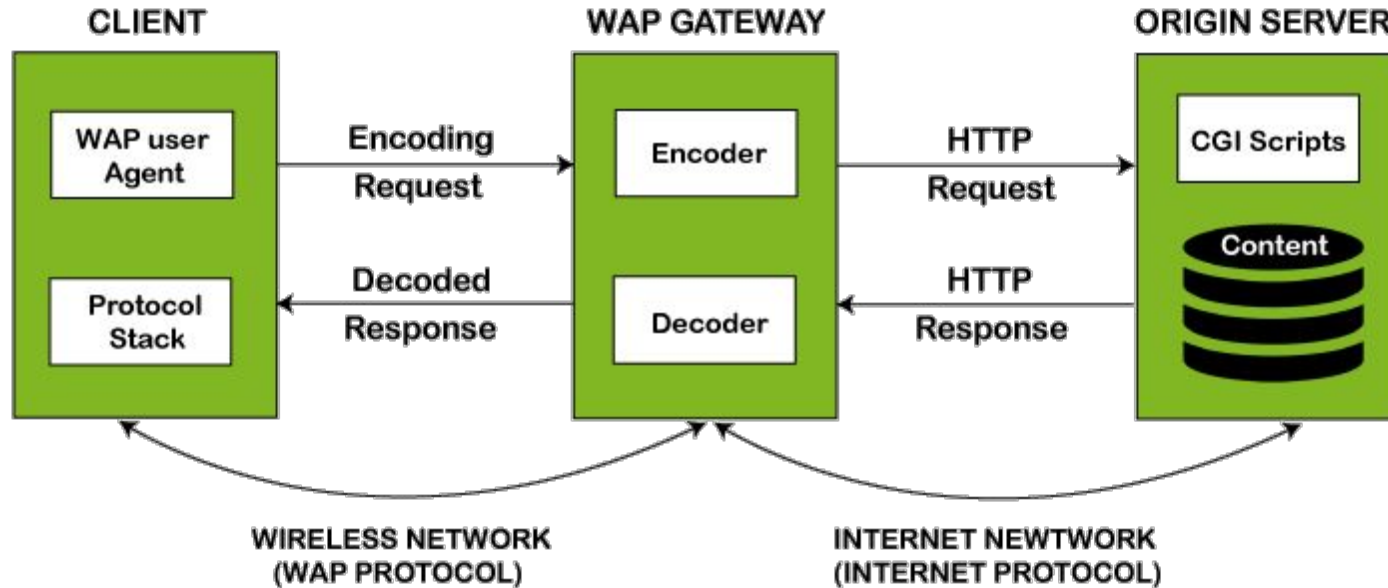
It enables creating web applications for mobile devices.

WAP

Wireless Application Protocol (WAP) is a technology that allows mobile phones to access the internet. It was mainly used in older phones before smartphones became popular. WAP helped users browse websites, send emails, and do online banking, even with slow internet connections. However, it is now outdated because modern smartphones use faster networks and better browsers.



The user opens the mini-browser in a mobile device. He selects a website that he wants to view. The mobile device sends the URL encoded request via network to a WAP gateway using WAP protocol.



The WAP gateway translates this WAP request into a conventional HTTP URL request and sends it over the internet. The request reaches to a specified Web server and it processes the request just as it would have processed any other request and sends the response back to the mobile device through WAP gateway in WML file which can be seen in the micro-browser.



This image shows an older model HTC smartphone with a slide-out QWERTY keyboard and a stylus.

The WAP standard described a protocol suite or stack allowing the interoperability of WAP equipment and software with different network technologies, such as GSM and IS-95 (also known as CDMA).



Wireless Application Environment (WAE)	WAP protocol suite
Wireless Session Protocol (WSP)	
Wireless Transaction Protocol (WTP)	
Wireless Transport Layer Security (WTLS)	
Wireless Datagram Protocol (WDP)	
*** Any wireless data network ***	



Mobile version of facebook
that got deprecated in 2012.
Could be accessed by
`m.facebook.com`



11% of Indonesians who said they used Facebook also said they did not use the Internet.

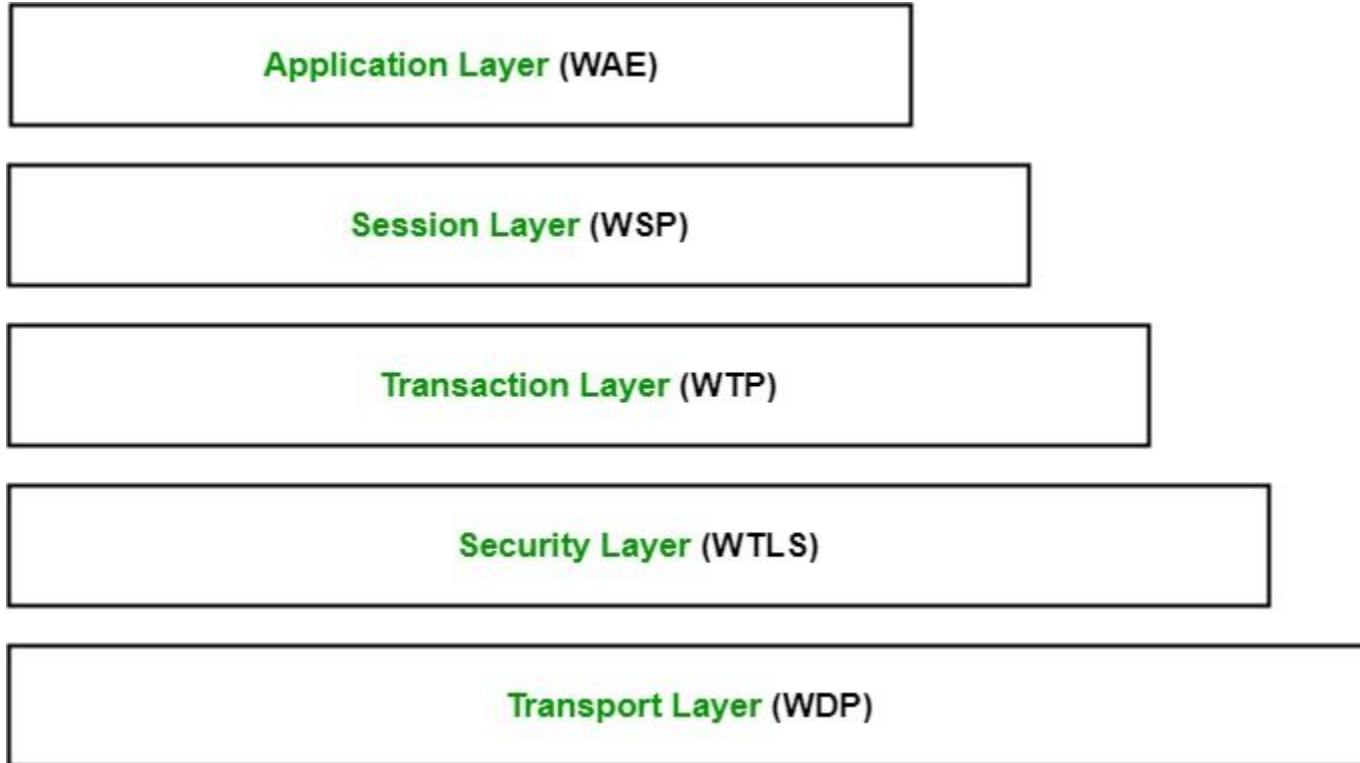
65% of Nigerians, 61% of Indonesians, and 58% of Indians agree with the statement that "Facebook is the Internet"

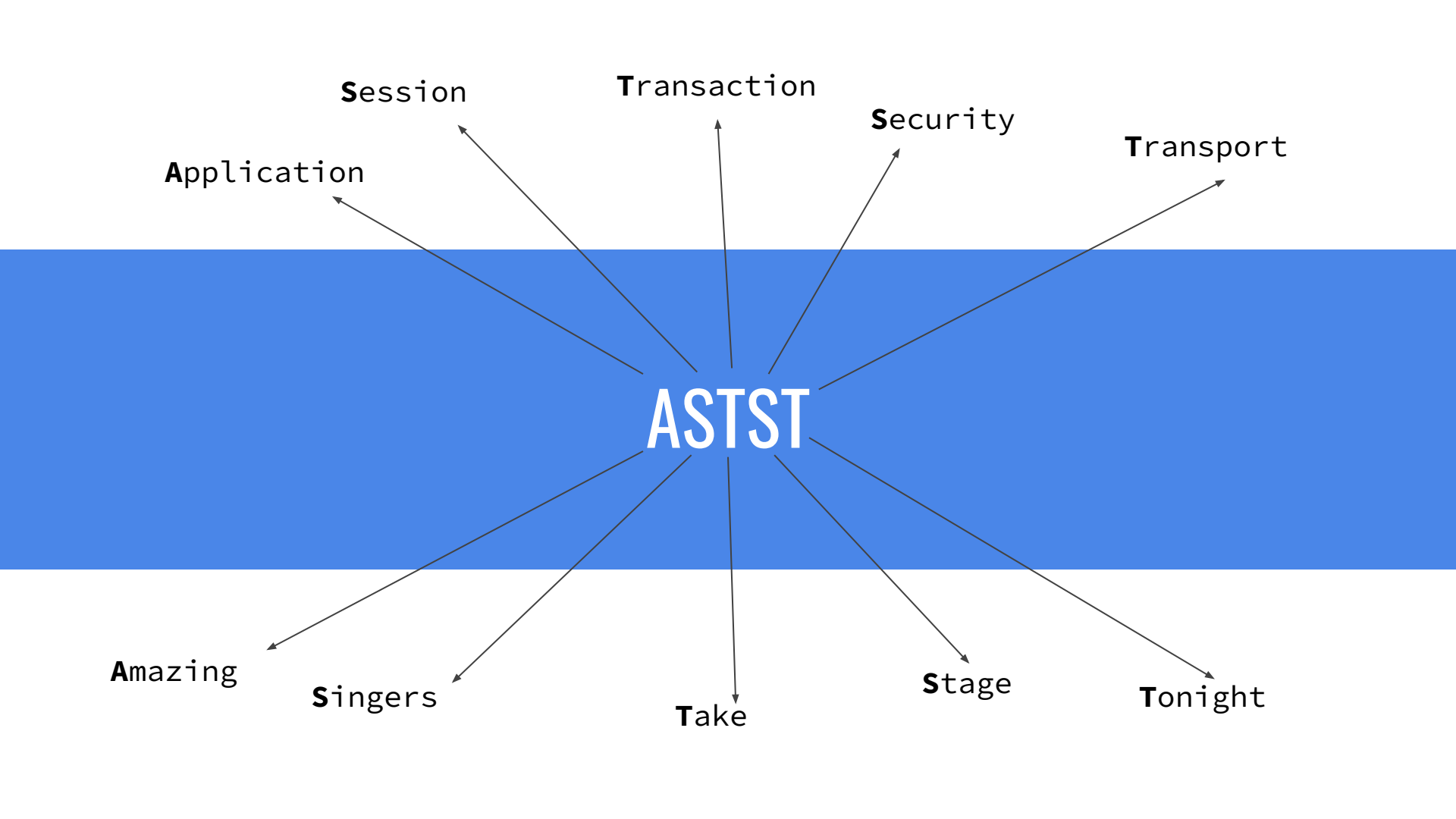
A woman with a joyful expression is laughing heartily. She is wearing a purple short-sleeved shirt with a colorful geometric pattern and a bright orange apron. She is positioned behind a market stall, with various items like a blue basket and a white bowl visible. In the background, other people and market structures are blurred, suggesting a busy outdoor market environment. The text "We build for the world" is overlaid in white on the center of the image.

We build for the world

Architecture of WAP

Architecture of WAP





ASTST

Application

Session

Transaction

Security

Transport

Tonight

Stage

Take

Singers

Amazing

1. Application Layer:
This layer contains the **Wireless Application Environment (WAE)**. It contains mobile device specifications and content development programming languages like WML.
2. Session Layer:
This layer contains **Wireless Session Protocol (WSP)**. It provides fast connection suspension and reconnection.
3. Transaction Layer:
This layer contains **Wireless Transaction Protocol (WTP)**. It runs on top of UDP (User Datagram Protocol) and is a part of TCP/IP and offers transaction support.
4. Security Layer:
This layer contains **Wireless Transaction Layer Security (WTLS)**. It offers data integrity, privacy and authentication.
5. Transport Layer:
This layer contains **Wireless Datagram Protocol**. It presents consistent data format to higher layers of WAP protocol stack.

Wireless Application Exchange

Definition: A framework that allows mobile-friendly apps and websites to run on older mobile devices.

Analogy: Like a simplified version of a website designed to work on small phone screens with limited capabilities.

Example: A mobile banking site that loads quickly on old mobile phones, using basic text and simple images.

Wireless Session Protocol

Definition: A protocol that helps manage and maintain communication between a mobile device and a server, similar to how HTTP works for websites.

Analogy: Like a waiter in a restaurant who takes your order to the kitchen and brings back your food, ensuring smooth communication.

Example: When you browse a news website on your phone, WSP keeps the connection active so you can read articles without interruptions.

Wireless Transaction Protocol

Definition: Ensures that important mobile transactions, like payments, are completed successfully or not at all, preventing half-finished operations.

Analogy: Like an online money transfer system that either fully sends the money or cancels the transaction—never leaving it halfway.

Example: When you buy a movie ticket online, WTP makes sure your money is deducted only if the ticket is successfully booked.

Wireless Transport Layer Security

Definition: A security protocol that encrypts and protects data being transmitted over wireless networks, preventing hacking or interception.

Analogy: Like sending a secret message in a locked box so only the intended recipient can open it.

Example: When you enter your password on a banking app, WTLS ensures that no one else can see or steal it.

Wireless Datagram Protocol

Definition: A protocol responsible for sending small packets of data over wireless networks without ensuring they are received.

Analogy: Like a pizza delivery person who drops off orders at the door but doesn't wait for confirmation from the customer.

Example: A weather app fetching live updates—WDP quickly sends the temperature data but doesn't wait for a reply.

Term	Nickname	What it does?	Example
WAE	Mobile-Friendly Internet	Helps websites/apps work on older mobile phones	Mobile banking site
WSP	The Messenger	Manages the connection between phone & website	Reading news online
WTP	The Deal Maker	Ensures transactions complete properly	Buying a movie ticket online
WTLS	The Bodyguard	Protects data from hackers	Secure password entry
WDP	The Delivery Guy	Sends small data packets	Weather updates on an app

Do you know ?

**Multimedia Messaging
Service (MMS)
is a combination of
WAP and SMS
allowing for sending
of picture messages.**

Working of WAP

Working mechanism of WAP

The user opens the mini-browser in a mobile device. He selects a website that he wants to view.

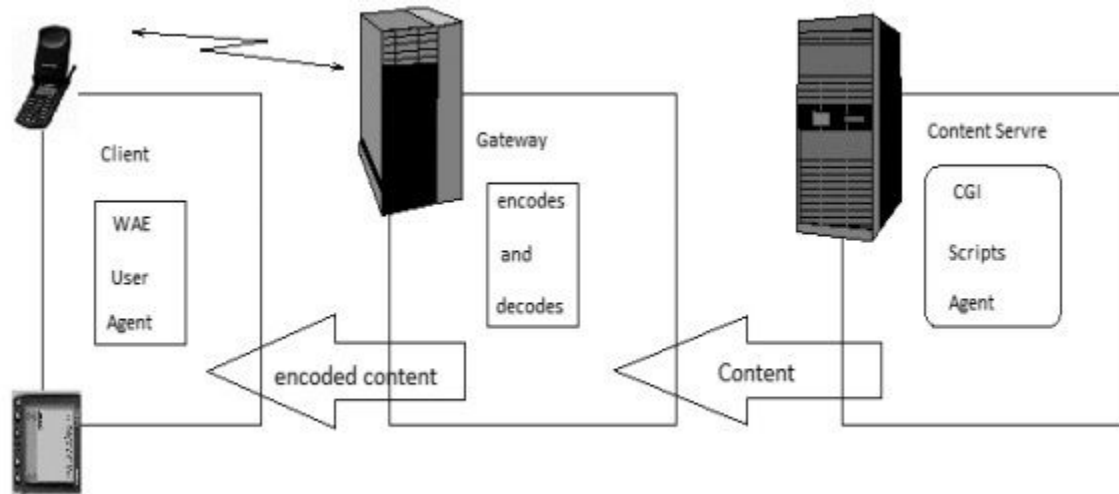
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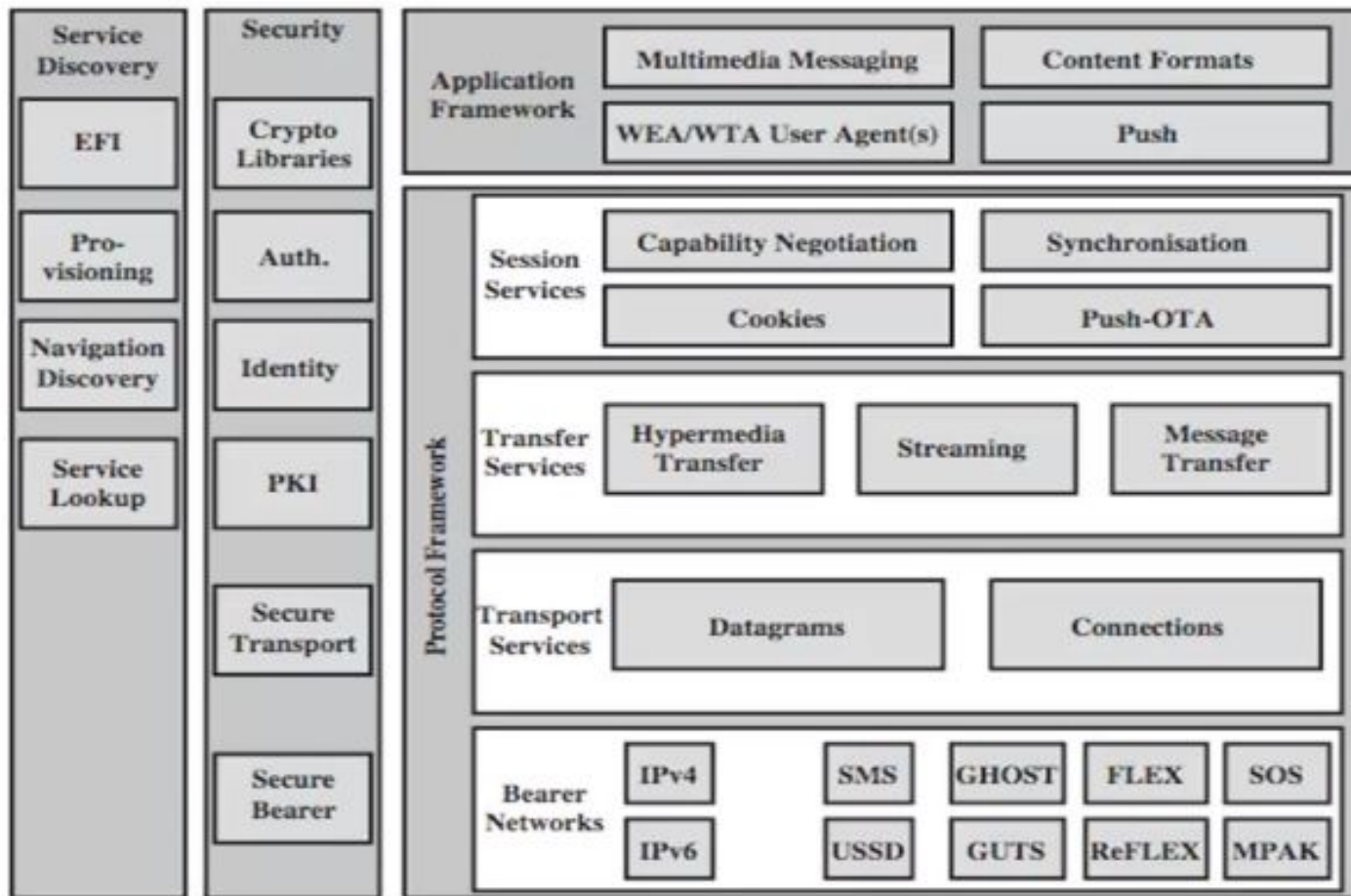
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The request reaches to a specified web servers and it process the request just as it would have processed any other request and sends the response back to the mobile device through WAP gateway in WML file which can be seen in the micro-browser.

Working of WAP

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WAP Architecture

WAP allows mobile users to access web-based content using limited bandwidth and small displays. The architecture is divided into different functional layers.

Service Discovery & Security

Helps users find and access mobile services (like browsing websites).

Ensures security with authentication and encryption (protects user data).

Uses PKI (Public Key Infrastructure) for secure communication.

Public Key Infrastructure (PKI) is a security system that helps protect online communication by using encryption and digital certificates. It ensures that data is secure, authentic, and only accessed by the right people.

Application Framework

Defines how WAP applications work on mobile devices.

Supports features like multimedia messaging, push notifications, and content formats.

Protocol Framework (Manages Data Transfer)

Session Services: Handles cookies, synchronization, and capability negotiation (adjusts content to fit mobile screens).

Transfer Services: Manages text, images, videos, and messages for smooth communication.

Transport Services: Ensures data is correctly sent and received using reliable connections.

Bearer Networks (Supports Different Mobile Networks)

Works with IPv4, IPv6 (internet communication protocols).

Supports SMS, USSD, and other mobile messaging services for communication.

Technology	Speed	Calling	Messaging	VAS
2G	 Download up to 236 Kbps; Upload up to 236 Kbps	 Voice calls	 Text messaging, Basic MMS	 Caller Ring Back Tunes, Ringtones, etc.
3G	 Download up to 21 Mbps; Upload up to 5.7 Mbps	 Video calls	 MMS Video Messages	 Video on Demand, Location Based Services and lots more

1G	2G	3G	4G	5G
Released: 1979 Standards: NMT, AMPS & TACS Capabilities: • Analog voice	Released: 1991 Standards: GSM & CDMA Capabilities: • Digital voice • Encrypted communication • Limited roaming • SMS & MMS Extensions: • GPRS (2.5G) • CDMA2000 (2.5G) • EDGE (2.75G)	Released: 2002 Standards: UMTS & EV-DO Capabilities: • Mobile broadband • Locating services • Multimedia streaming • Seamless global roaming Extensions: • HSPA+ (3.5G)	Released: 2009 Standards: LTE Capabilities: • High Speed mobile Internet • IP-based packet switching • HD multimedia streaming • Seamless global roaming Extensions: • Feature extension through new category/releases	Released: 2019 Standards: 5G Capabilities: • Private networks (local use frequency) • (l)IoT Ready • Massive Machine Type communication • Ultra-low-latency • Ultra-high reliability • Millimeter wave support Extensions: • Feature extension through new categories/releases
 0.0024 Mbit/s	 0.064 Mbit/s	 42 Mbit/s	 1000 Mbit/s	 10,000 Mbit/s
Industry Impact: – • No impact on industrial applications	Industry Impact: 0 • Remote control / Telecontrol • Text messages from and to remote machines	Industry Impact: + • Video monitoring • Remote Access to machines (e.g. for teleservice) • Remote Condition Monitoring	Industry Impact: ++ • Mobile service Technicians • Service via smart phones • Wireless Backhaul	Industry Impact: +++ • Autonomous Logistics • Autonomous Machines • Assisted Work • Wireless Backhaul • Edge Computing • Mobile Equipment

Challenges

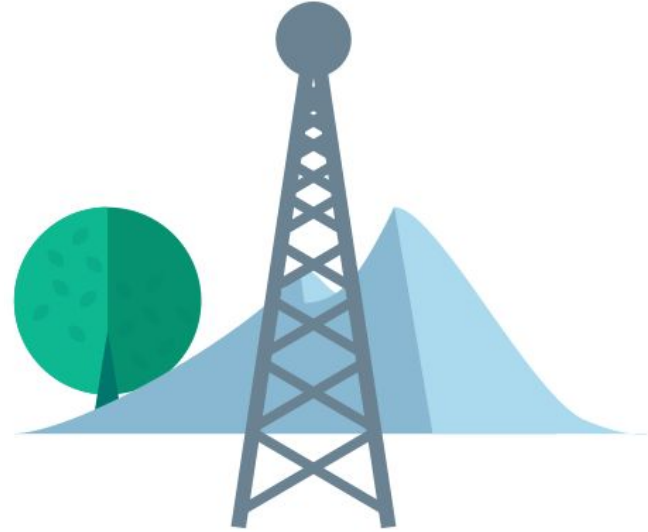
- Support to the application with huge content and graphics
- Assurance of Service
- Bandwidth
- Device Capability
- Usage of network
- Scalability of Markup Language
- Interface design for handheld devices



Summary

— — —

- WAP enables wireless device to use Internet
- Specific need and requirement for wireless devices
- Limited device capability is always a major concern
- WAP supports all major networks CDMA, GSM etc.
- WAP supports almost all operating system Windows, Java based etc.





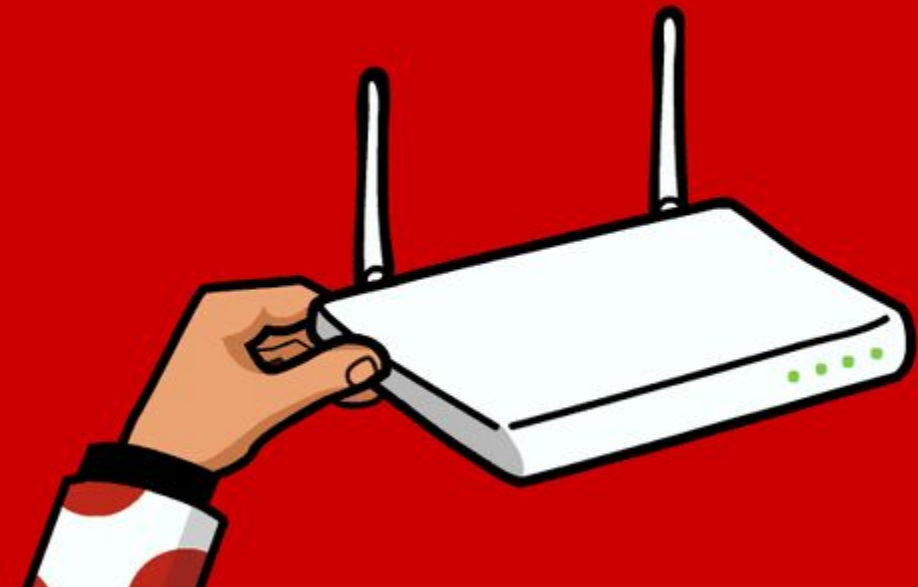


Unit 2

Wireless Technologies

Wireless technology refers to the transmission of data, voice, and signals without using physical cables or wires.

Examples include Wi-Fi, Bluetooth & mobile networks (4G, 5G) etc..



- ADSL
- WiMAX
- WLAN
- WMAN
- Wi-Fi

ADSL

Asymmetric Digital Subscriber Line

Asymmetric Digital Subscriber Line

— — —

A data communications technology that enables faster data transmission over copper telephone lines than a conventional voiceband modem



ADSL

ADSL is a high speed internet-access service that utilizes existing copper telephones lines to send and receive data at speeds that far exceeds conventional dial-up modems.

ADSL uses standard telephone lines to transmit upstream and downstream data on a digital frequency, which sets these data streams apart from the analog signals telephone and fax machines use.

ADSL allows data stream speeds from 1.5 to 8 megabits per second (Mbps).

ADSL download/ upload

The “asymmetric” in ADSL refers to the fact that the downstream data rate, or the data coming to your computer from the internet, is travelling faster than the upstream data, or the data travelling from your computer to the Internet.

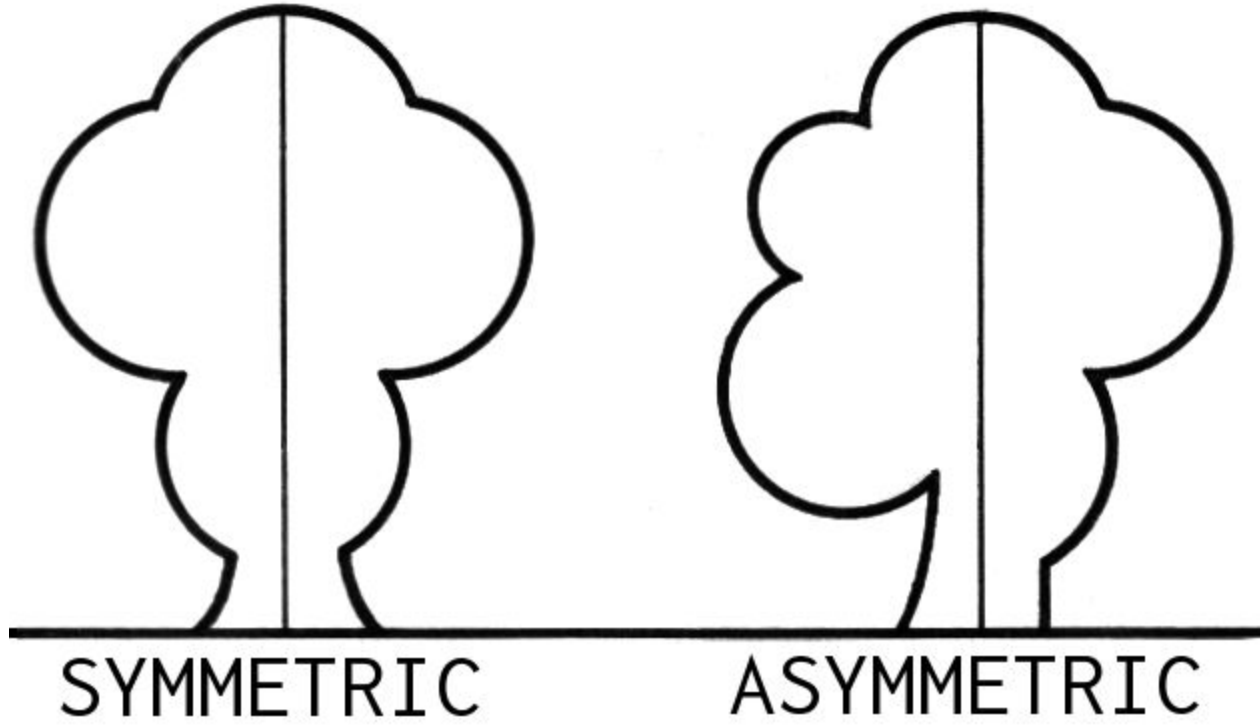
Upstream data rates are slower because Web page requests are fairly miniscule data strings that don't require much bandwidth to handle efficiently.

ADSL भनेको इन्टरनेट जडान गर्ने प्रविधि हो जसले परम्परागत तामाको टेलिफोन तारहरू (Copper Wires) मार्फत उच्च गतिको डाटा ट्रान्समिसन सम्भव बनाउँछ।

ADSL प्रयोग गर्दा टेलिफोन एक्सचेन्जको दूरी महत्वपूर्ण हुन्छ, किनभने दूरी बढ्दै जाँदा इन्टरनेट गति कम हुन्छ। यो असिमेट्रिक भएकाले डाउनलोड स्पीड अपलोड स्पीडभन्दा छिटो हुन्छ। टेलिफोन र इन्टरनेट एकसाथ चलाउन स्प्लिटर प्रयोग गर्नुपर्छ। साथै, यो सधैं जडान रहने भएकाले सुरक्षाका उपायहरू अपनाउनुपर्छ।

Note

स्प्लिटर भनेको एउटा सानो यन्त्र हो, जसले टेलिफोन लाइनबाट आउने सिग्नललाई दुई भागमा छुट्याउँछ— एक भाग टेलिफोनको लागि र अर्को भाग इन्टरनेट (ADSL मोडेम) को लागि।

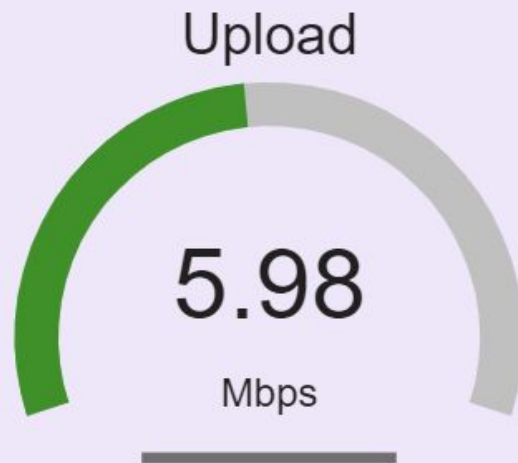
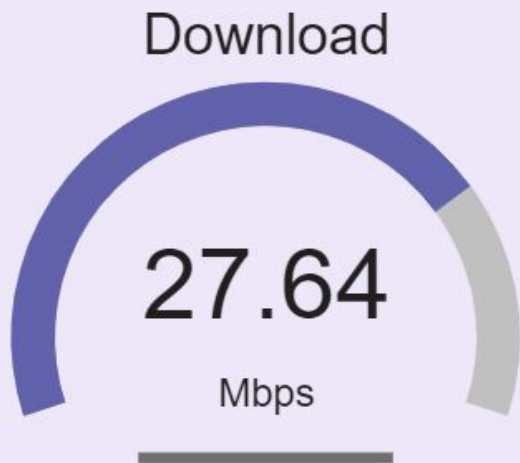


डाउनलोड = अपलोड

डाउनलोड > अपलोड

Asymmetric Digital Subscriber Line

Bandwidth and bit rate are said to be asymmetric, meaning greater toward the customer premises (downstream) than the reverse (upstream).



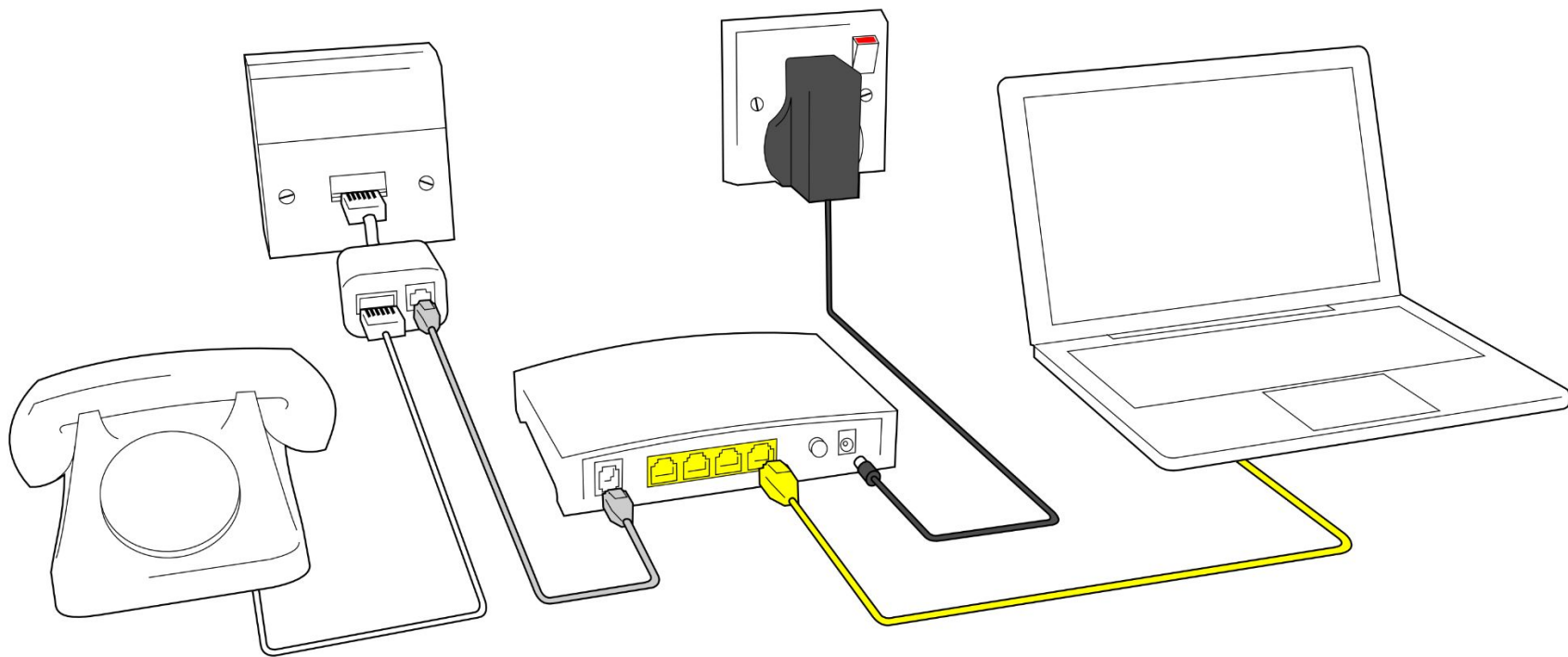
ADSL provides asymmetric data transfer, meaning download speeds are higher than upload speeds, which is suitable for general internet usage like browsing, streaming, and downloading.

How it works

ADSL works by using spectrum above the band used by voice telephone calls.

With a DSL filter, often called splitter, the frequency bands are isolated, permitting a single telephone line to be used for both ADSL service and telephone calls at the same time.





A typical ADSL setup

Components in the image ...

Telephone Line Connection

ADSL Splitter/Filter

ADSL Modem/Router

Computer Connection

The setup starts with a telephone wall socket that provides the main telephone line.

The line is split into two connections: one for the telephone and the other for the ADSL modem.

A splitter or microfilter is connected to the telephone line.

It separates low-frequency signals (for voice calls) and high-frequency signals (for internet data) to prevent interference.

The ADSL modem/router is connected to the telephone line through the splitter.

It converts the ADSL signal into digital data for internet access.

The modem has multiple Ethernet ports to connect devices like laptops, desktop computers, or routers.

A laptop is connected to the modem via an Ethernet cable (yellow cable in the image).

This allows internet access by transmitting data between the modem and the computer.

ADSL Standards

Version	Standard name	Common name	Downstream rate	Upstream rate	Approved in
ADSL	ANSI T1.413-1998 Issue 2	ADSL	8.0 Mbit/s	1.0 Mbit/s	1998
	ITU G.992.2	ADSL Lite (G.lite)	1.5 Mbit/s	0.5 Mbit/s	1999-07
	ITU G.992.1	ADSL (G.dmt)	8.0 Mbit/s	1.3 Mbit/s	1999-07
	ITU G.992.1 Annex A	ADSL over POTS	12.0 Mbit/s	1.3 Mbit/s	2001
	ITU G.992.1 Annex B	ADSL over ISDN	12.0 Mbit/s	1.8 Mbit/s	2005
ADSL2	ITU G.992.3 Annex L	RE-ADSL2	5.0 Mbit/s	0.8 Mbit/s	2002-07
	ITU G.992.3	ADSL2	12.0 Mbit/s	1.3 Mbit/s	2002-07
	ITU G.992.3 Annex J	ADSL2	12.0 Mbit/s	3.5 Mbit/s	2002-07
	ITU G.992.4	Splitterless ADSL2	1.5 Mbit/s	0.5 Mbit/s	2002-07
ADSL2+	ITU G.992.5	ADSL2+	24.0 Mbit/s	1.4 Mbit/s	2003-05
	ITU G.992.5 Annex M	ADSL2+M	24.0 Mbit/s	3.3 Mbit/s	2008

Advantages



- Internet is always on and the phone line is free for voice calls if needed
- The speed is much higher than a regular modem
- DSL doesn't necessarily require new wiring
- The company that offers DSL will usually provide the modem as part of the installation.
- Since it is Digital, conversion is not necessary.



Disadvantages

- ADSL connection works better when you are closer to the provider's central office. The farther away you get from the central office, the weaker the signal becomes.
- The connection is faster for receiving data than it is for sending data over the Internet.
- The service is not available everywhere.

WiMAX

(Worldwide Interoperability for Microwave Access)

WiMAX

WiMAX (Worldwide Interoperability for Microwave Access) is a family of wireless broadband communication standards based on the IEEE 802.16 set of standards, which provide multiple physical layer (PHY) and Media Access Control (MAC) options.



WiMAX, which will have a range of up to 31 miles, is primarily aimed at making broadband network access widely available.

WiMAX is a wireless communication technology that provides high-speed internet access over long distances. It is similar to Wifi but has a much wider coverage area and can be used as an alternative to wired broadband connections like ADSL or fiber.

WiMAX is a wireless communication technology that provides high-speed internet service over a large area. It is primarily used for internet distribution in rural and remote areas.

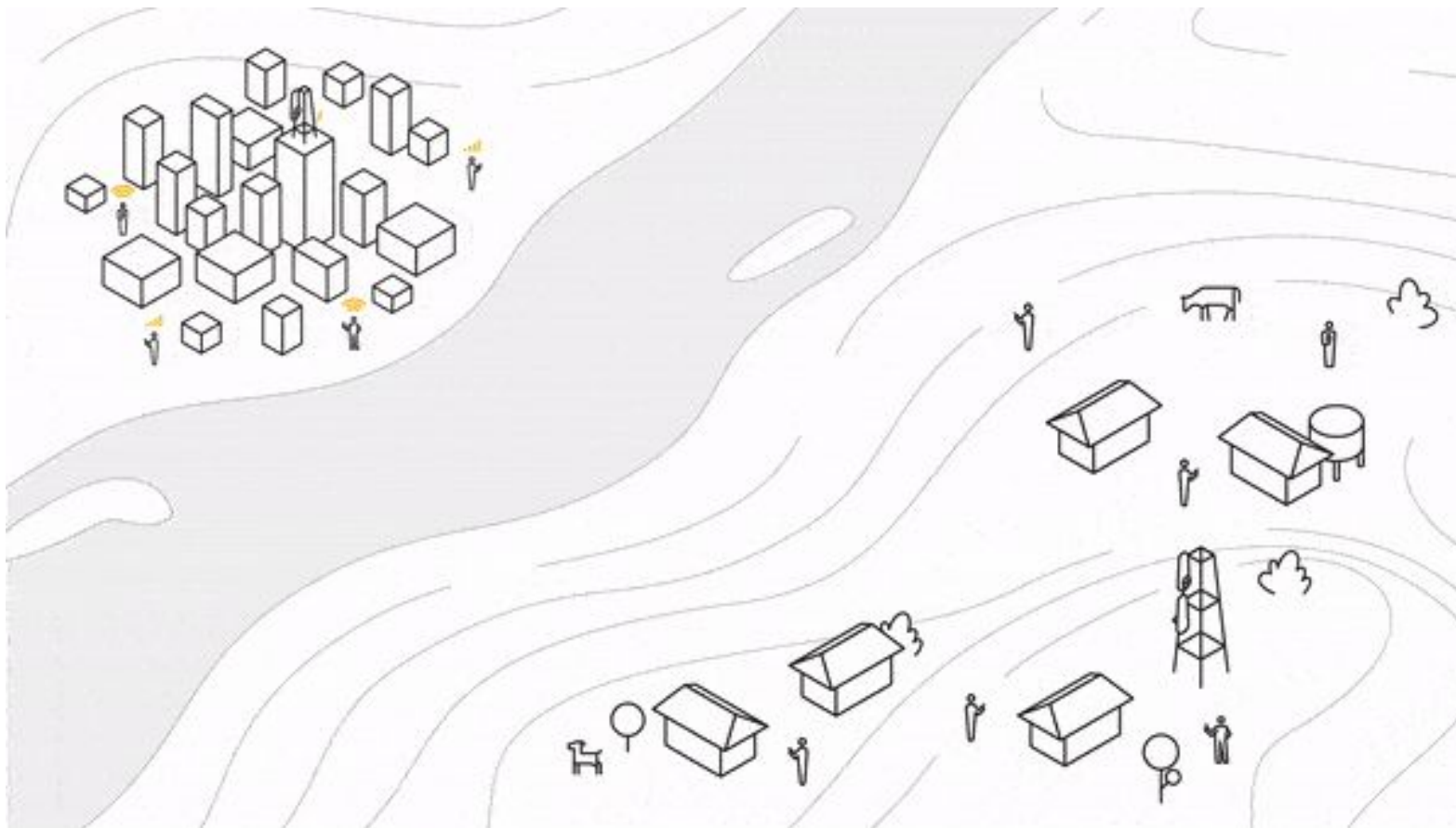
WiMAX एक बेतार सञ्चार प्रविधि हो, जसले
ठूलो भू-भागमा उच्च गतिको इन्टरनेट सेवा प्रदान
गर्छ। यसलाई मुख्य रूपमा ग्रामीण तथा दुर्गम
क्षेत्रमा इन्टरनेट वितरण गर्न प्रयोग गरिन्छ।

WiMAX

WiMAX stands for Worldwide Interoperability for Microwave Access.

Orthogonal Frequency Division Multiplexing (OFDM) (carriers of width of 5 MHz or greater can be used) connectivity at speeds up to 70 Mbps.

WiMAX refers to broadband wireless networks that are based on the IEEE 802.16 standard, which ensures compatibility and interoperability between broadband wireless access equipment.



Wireless broadband access as an alternative to cable and DSL.

HOW WiMAX WORKS



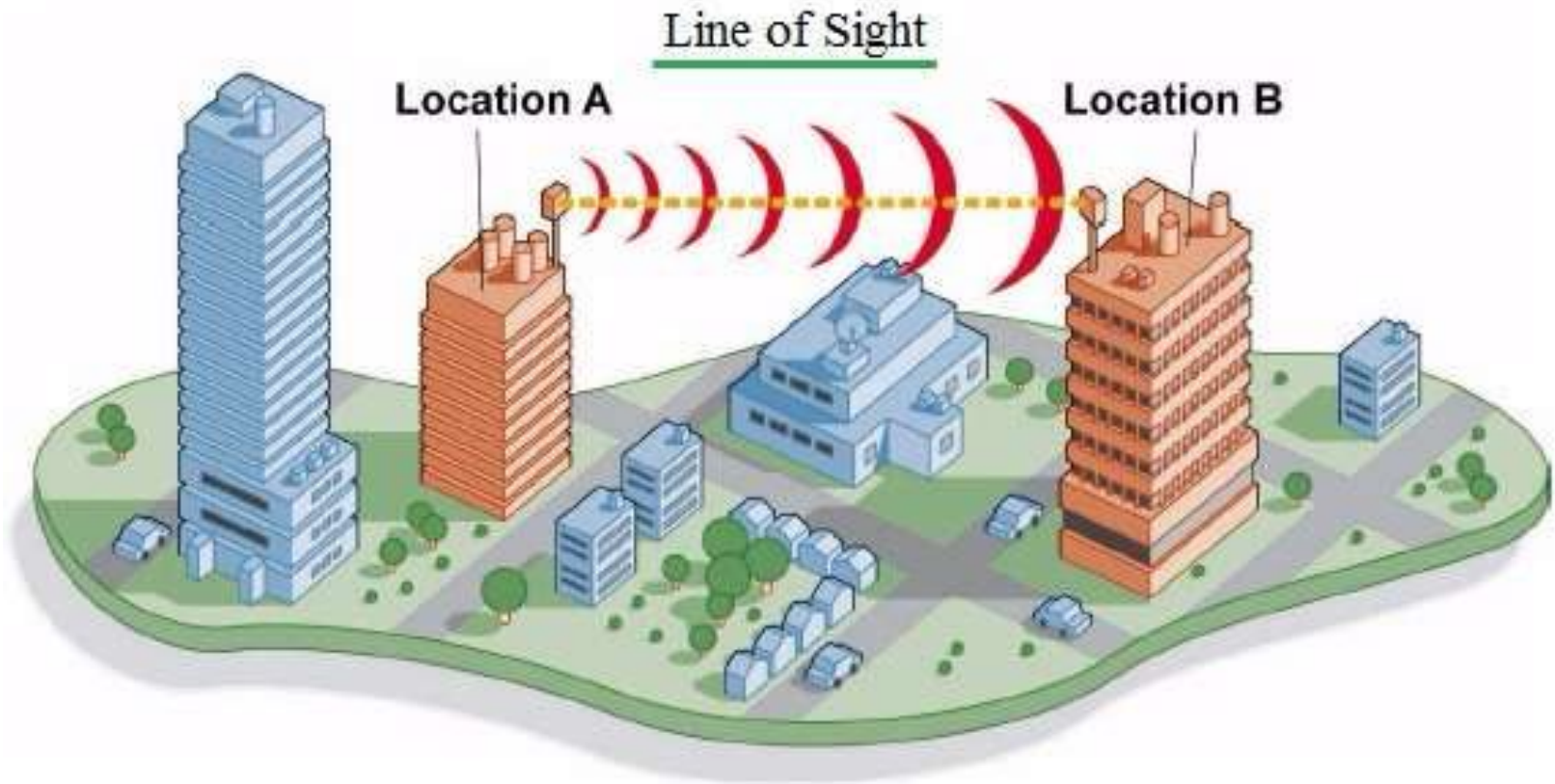
A WiMAX system consists of:

WiMAX receiver:

- The receiver could be a small box or PCMCIA card, or they could be built into a laptop the way Wi-Fi access is today.

WiMAX tower:

- It can provide cover to a very large area as big as 3,000 square miles



Line-of-sight connection is stronger and more stable, so it's able to send a lot of data with fewer errors. LoS use higher frequencies, of around 66 GHz.



In this mode, WiMAX uses a lower frequency range
~ 2 GHz to 11 GHz (similar to WiFi).

Working of WiMAX

Signal Transmission: The base station transmits data using radio signals in the microwave frequency range (typically between 2.3 GHz and 3.5 GHz).

Connection to Users:

Subscriber stations receive the signal, and data is transferred between the base station and the user's device.

Data Processing & Internet

Access: The base station connects to an internet gateway, enabling users to browse the internet, stream videos, and perform other online activities.

Advantages



- WiMAX 802.16 is surely an advanced technology when compared with the other technologies with more
 - Bandwidth
 - Bit rate
 - Coverage area
 - Cable cost reduced
 - Performance
 - Quality of service and secure too



Disadvantages

- Interference with Bluetooth wireless communication
- The cost of equipment is really high and it has not yet widely used.
- Power-consuming technology
- High installation costs
- Signal interruption in bad weather

Classwork : Write down any five key features of
WiMAX

Key Features of WiMAX

— — —

High-Speed Internet: Supports data rates up to 1 Gbps (for fixed applications).

Wide Coverage: Can cover up to 50 km (31 miles) for fixed stations and 5-15 km (3-10 miles) for mobile users.

Wireless Connectivity: Eliminates the need for physical cables, making deployment easier in remote areas.

Multiple Access Modes: Supports both fixed and mobile wireless broadband.

Security: Uses strong encryption and authentication protocols.

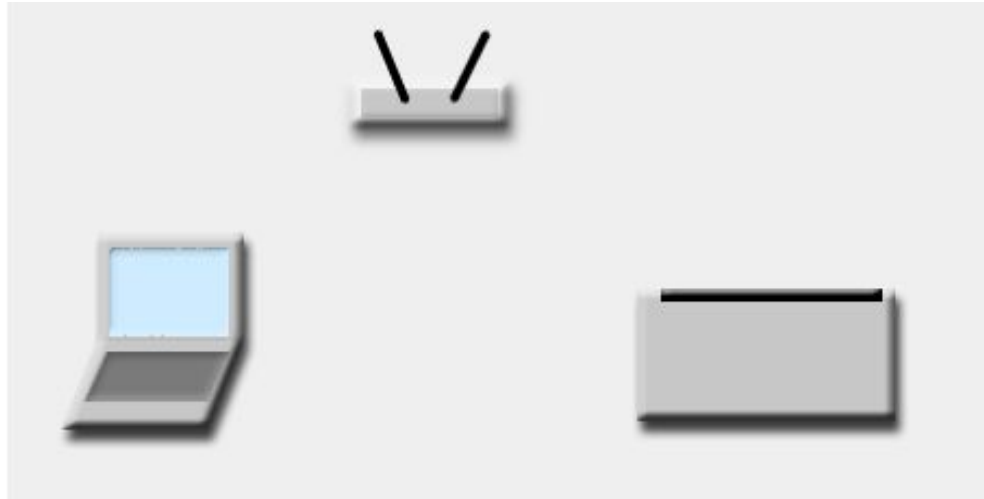
WLAN

Wireless LAN

Wireless LAN (WLAN)

A wireless LAN (WLAN) is a wireless computer network that links two or more devices using wireless communication to form a local area network (LAN) within a limited area.

Through a gateway,
a WLAN can also provide
a connection to the
wider Internet.



WLAN is a wireless network that connects devices within a limited area, such as a home, office, or school, using Wi-Fi or other wireless communication technologies.

Advantages & Disadvantages of WLAN

Advantages:

No physical cables required

Easy to expand and set up

Provides mobility within a specific range

Disadvantages:

Limited range (usually up to 100 meters)

Susceptible to interference from other devices

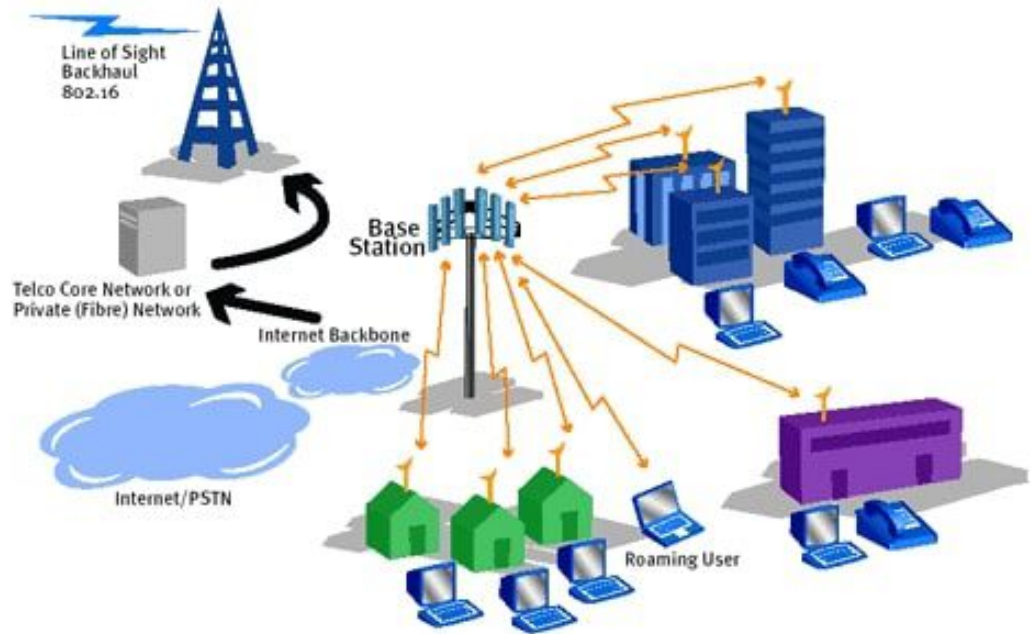
Security concerns (hacking, unauthorized access)

WMAN

WirelessMAN

Wireless Metropolitan Area Network (WMAN)

Wireless Metropolitan Area Network (WMAN) is a type of Metropolitan Area Network (MAN) where the users are connected wirelessly over a large area.



WMAN is a wireless network that covers a metropolitan area, such as a city or large campus, providing high-speed connectivity over a wider range than WLAN.

Examples include:

WiMAX (Worldwide Interoperability for
Microwave Access)

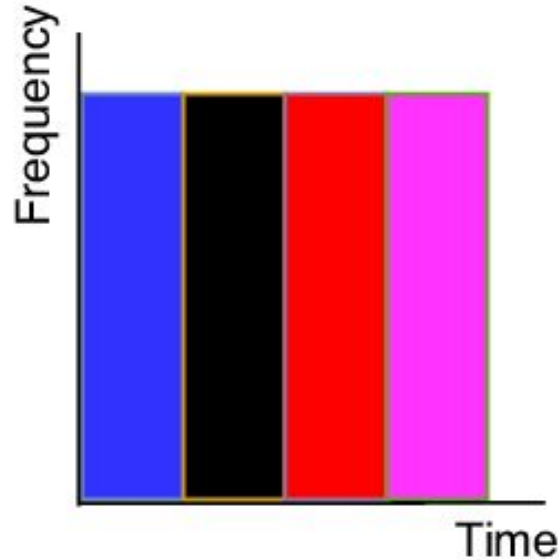
4G LTE networks covering cities

Public Wi-Fi hotspots across a
metropolitan area

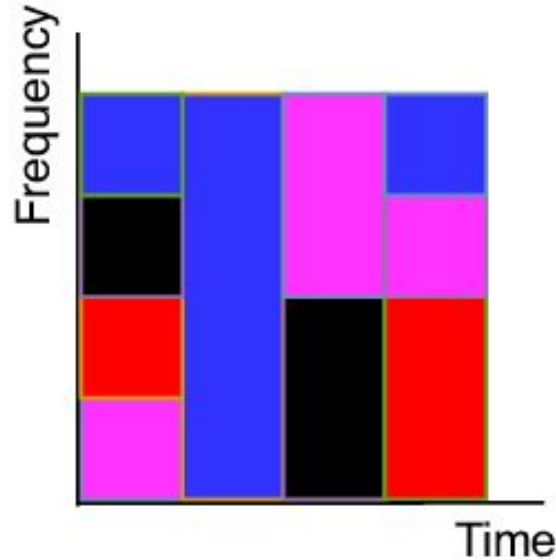
Characteristics of WMAN

1. Connection can be Point to Point or Point to Multipoint networks.
2. Service to multiple nodes from one access point.
3. Covers a larger area within a radius up to 50 km.
4. Stable connections to the terminals.

Orthogonal Frequency-Division Multiple Access



(a) OFDM



(b) OFDMA





WiFi 5 OFDM

Bandwidth wasted - Higher latency



WiFi 6 OFDMA

Bandwidth fully utilized - Lower latency

Benefits of WMAN

- Covers multiple locations within a metropolitan area.
- Does not require high cost for infrastructure in placing fiber or copper cabling and leasing lines.
- Works as backups for wired networks.
- Easy to use, extend, exchange.

Advantages & Disadvantages

— — —

Advantages:

Covers a large area (up to several kilometers)

Can be used for broadband internet access

Supports high-speed connectivity

Disadvantages:

Expensive infrastructure

Signal degradation due to weather and obstacles

Security risks in public networks

Wi-Fi

Wireless Fidelity

Wi-Fi wireless network protocols

- Based on the IEEE 802.11 family of standards
- Allows nearby digital devices to exchange data by radio waves.
- Most widely used computer network in the world



What exactly a wifi is?

Wireless Fidelity

Wireless technology that uses radio frequency to transmit data.

Vic-Hayes has been called the “father of wifi”

Wireless way to handle networking

Wi-Fi is a wireless networking technology that allows devices to connect to the internet or a local network using radio signals, typically in the 2.4 GHz and 5 GHz frequency bands.

Examples include:

Home Wifi routers

Public Wi-Fi hotspots

Corporate wireless networks

Privacy and Security

- MAC Address Filtering
- WEP (Wired Equivalent Privacy)
- WPA (Wi-Fi Protected Access)
- WPA2 (Wi-Fi Protected Access II)
- WPA3 (Wi-Fi Protected Access III)



Homework...

Explain and write MAC Address

Filtering, WEP, WPA, WPA2, WPA3

Advantages & Disadvantages

— — —

Advantages:

Convenient and easy to use

High-speed internet access without cables

Supports multiple devices

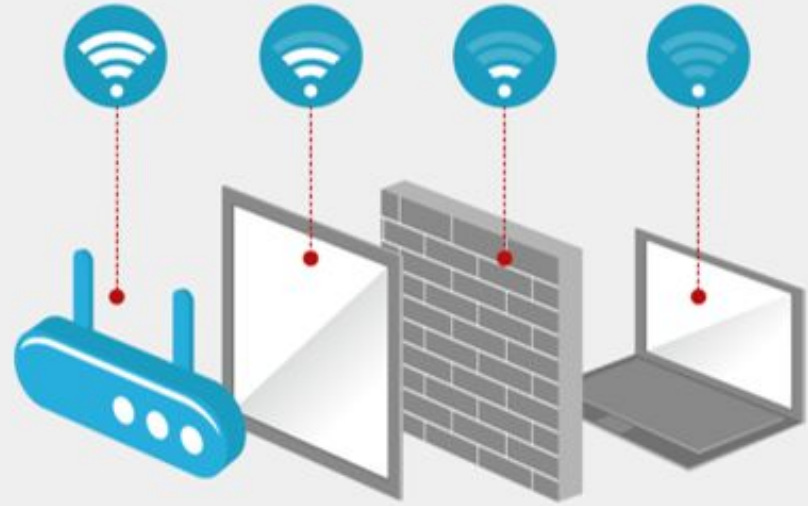
Disadvantages:

Limited range (~30–100 meters)

Performance drops with distance and interference

Security vulnerabilities if not properly configured

Many common obstructions such as walls, pillars, home appliances, etc. may greatly reduce range, but this also helps minimize interference between different networks in crowded environments.





Wirefree router
(no cable connection
needed)



**WiFi data
plans**



NEPAL TELECOM

3G WIFI ROUTER

Mobile data
Off



Auto-rotate



Hotspot



Battery Saver



Airplane



NT WiFi @ just Rs.10



NT Wifi Hotspots is now available @ Rs 10 per hour with the validity of 24 hrs.

Type **Wifi** and Send to **1416** to get Password

NT Notice Board For WiFi Hotspots:

1 6 1 8 0 7 0 7 4 9 4 3 4



खिटो मनेकै
WORLD**LINK**



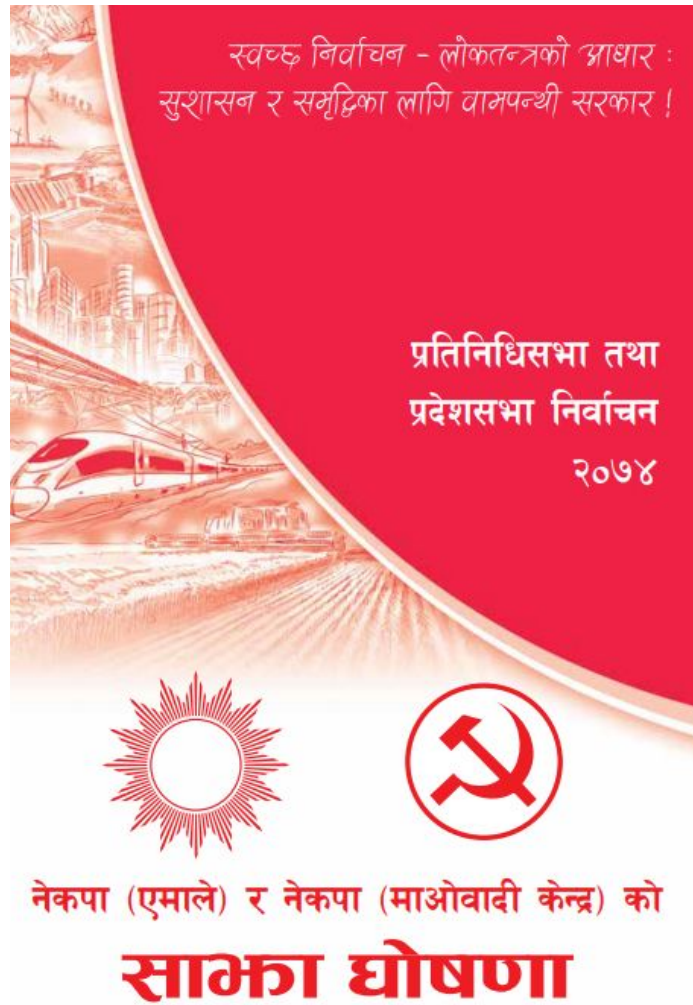
express**wifi**
by **facebook**



95,693

Free WiFi's in Nepal

Connect with the best WiFi hotspots in Nepal

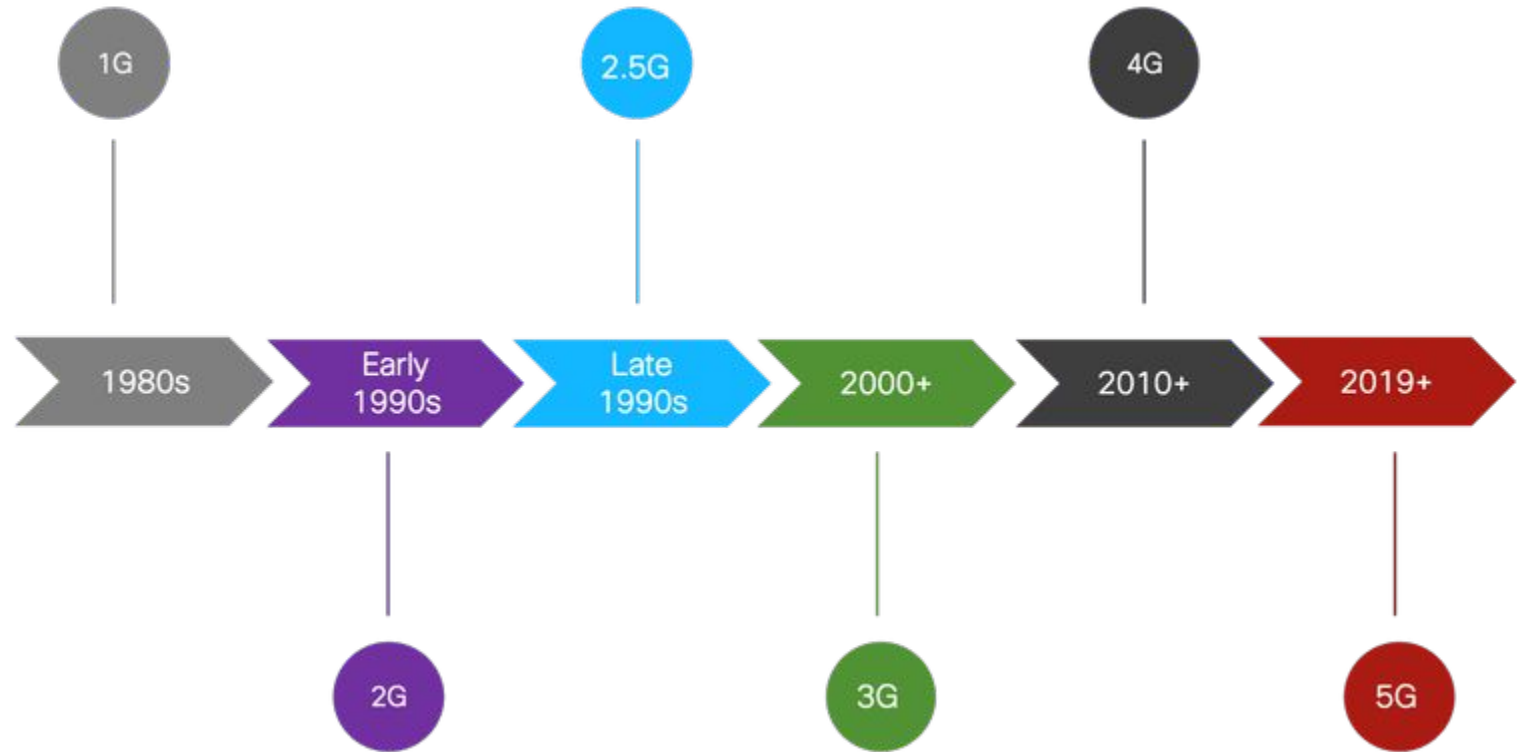


सबै विद्यालय, प्रशासनिक भवन, सभाभवन, स्वास्थ्य चौकी, हुलाक कार्यालय र सेवाकेन्द्रमा निःशुल्क इन्टरनेट सुविधा उपलब्ध गराइनेछ । आगामी पाँच वर्षमा देशभरिका महत्वपूर्ण सार्वजनिक स्थलहरूमा वाइफाई सेवा निःशुल्क गराइनेछ ।



- UMTS (3G)

- LTE (4G)



US CELL PHONE EVOLUTION

HOW WE GOT FROM **HERE** TO **HERE**





UMTS 3G

Universal Mobile Telecommunications System



UMTS is a complete system architecture

- As in GSM emphasis on standardized interfaces
 - * mix and match equipment from various vendors
- Simple evolution from GPRS - allows one to reuse/upgrade some of the GPRS backhaul equipment
- Backward compatible handsets and signaling to support intermode and intersystem handoffs
 - * Intermode; TDD to FDD, FDD to TDD
 - * Intersystem: UMTS to GSM or UMTS to GPRS
- UMTS supports a variety of user data rates and both packet and circuit switched services

UMTS

3G, or third-generation mobile network technology, refers to the Universal Mobile Telecommunications System (UMTS), which is a major improvement over 2G (GSM and CDMA) networks. UMTS is a part of the International Telecommunication Union's IMT-2000 standard and provides higher data speeds, better voice quality, and support for mobile internet access.



UMTS is a third-generation (3G) mobile communication system that provides voice, text, and data services, using WCDMA (Wideband Code Division Multiple Access) technology.

UMTS architecture

UTRAN (UMTS Terrestrial Radio Access Network)

- Radio Network Subsystem (RNS)



UE (User Equipment)

CN (Core Network)

Advantages



- Supports data speeds up to 42 Mbps with HSPA+, enabling faster web browsing, video streaming, and file downloads.
- Provides better voice clarity and fewer dropped calls compared to 2G networks.
- Compatible with many international networks, allowing users to stay connected while traveling.
- Enables video calls, mobile TV, and online gaming.



Disadvantages

- 3G devices consume more battery than 2G due to higher processing and data transfer rates.
- Requires significant investment in base stations, network towers, and core infrastructure upgrades.
- While widely available, some rural areas may still have weak or no 3G coverage.
- Although much faster than 2G, it is significantly slower than 4G LTE and 5G, making it less suitable for modern applications.

LTE

Long Term Evolution



LTE is often referred to as Evolved Packet System (EPS)

3G



When you've been home for 3 hours
and you notice the 4G symbol
instead of the WIFI symbol



LTE (Long-Term Evolution) – 4G

LTE is a 4G technology that provides high-speed mobile internet, offering faster speeds for data-heavy activities like streaming videos and online gaming.

Example: Streaming HD videos or playing online games on your smartphone with 4G LTE.

Advantages

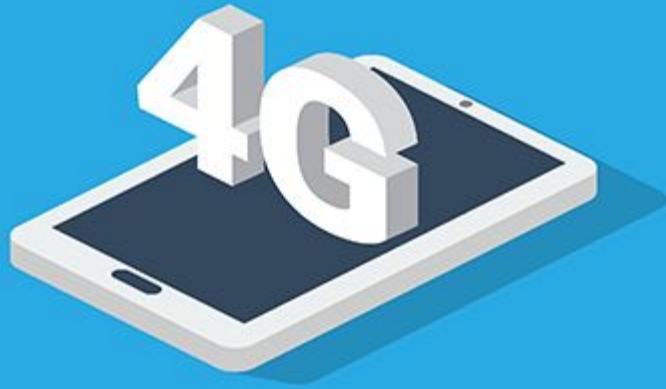


- Much faster speeds than 3G, allowing for high-definition video, faster downloads, and smoother gaming.
- Widely available around the world.



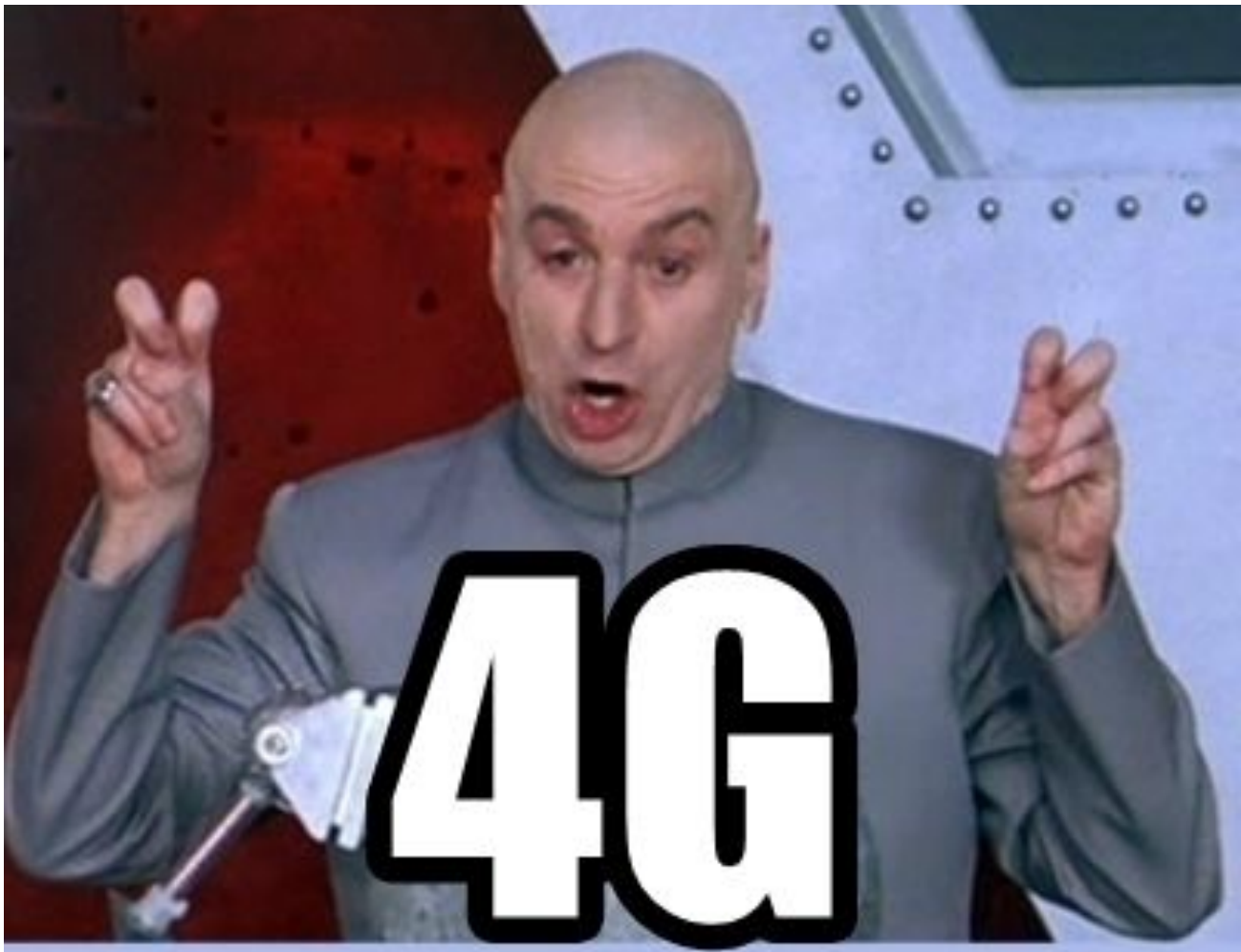
Disadvantages

- Can be expensive in some areas, depending on the service provider.
- Coverage can be spotty in rural areas or places with poor infrastructure.



Due to marketing pressures and the significant advancements that WiMAX, Evolved High Speed Packet Access, and LTE bring to the original 3G technologies, ITU later decided that LTE together with the aforementioned technologies can be called 4G technologies.

— — —



The image shows a stylized '4G' logo in a glowing blue font. The logo is positioned on the left side of the frame, with several horizontal blue light trails and small white dots extending to the right, creating a sense of motion and digital connectivity. The background is dark blue with a subtle gradient.

The LTE Advanced standard formally satisfies the ITU-R requirements to be considered IMT-Advanced.

To differentiate LTE Advanced and WiMAX-Advanced from current 4G technologies, ITU has defined them as "True 4G".

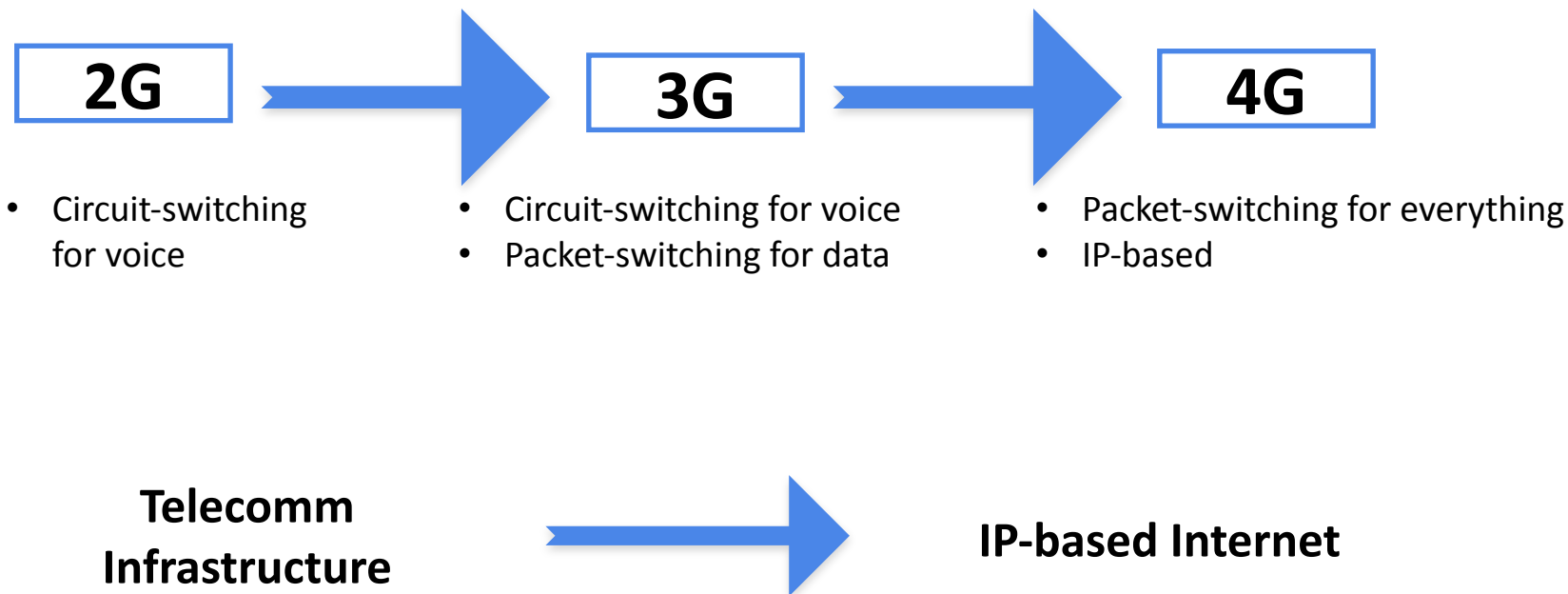
4G LTE: First global standard

- Increased speed
- IP-based network
- New air interface:
 - * OFDMA (Orthogonal Frequency-Division Multiple Access),
 - * MIMO (multiple antennas)
- Also includes duplexing, timing, carrier spacing, coding...
- New service paradigm (e.g., VoLTE)



Network Architecture Evolution

— — —



Multiple Input Multiple Output (MIMO)

Uses multiple transmitter and receiver antennas, which allow independent channels to be created in space.



LTE Network

Packet Delivery
Network Gateway (PGW)

- Connects LTE network to IP networks

Serving Gateway (SGW)

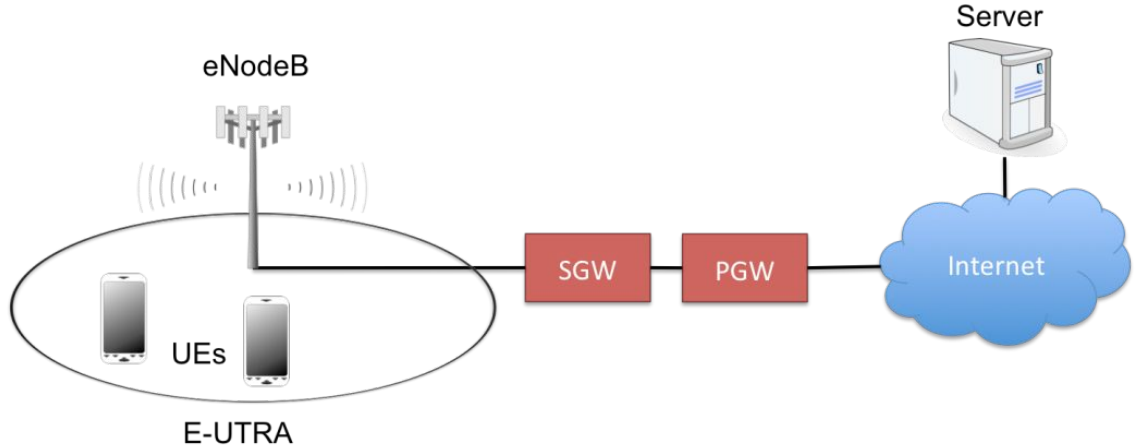
- Route packets to and from wireless access points

Enhanced Node B (eNodeB)

- Wireless access point

User Equipment (UE)

- End user devices



Advantages of LTE

- High throughput: High data rates (both downlink/uplink)
- Low latency: High speed network connections
- FDD and TDD in the same platform
- Superior end-user experience
- Seamless Connection: Existing networks supported
- Plug and play: No manual installation of drivers
- Simple architecture: Because of Simple architecture low operating expenditure (OPEX)



No 3G

No 4G

Only

Shiv G



5G NR New Radio





4G

low latency

5G

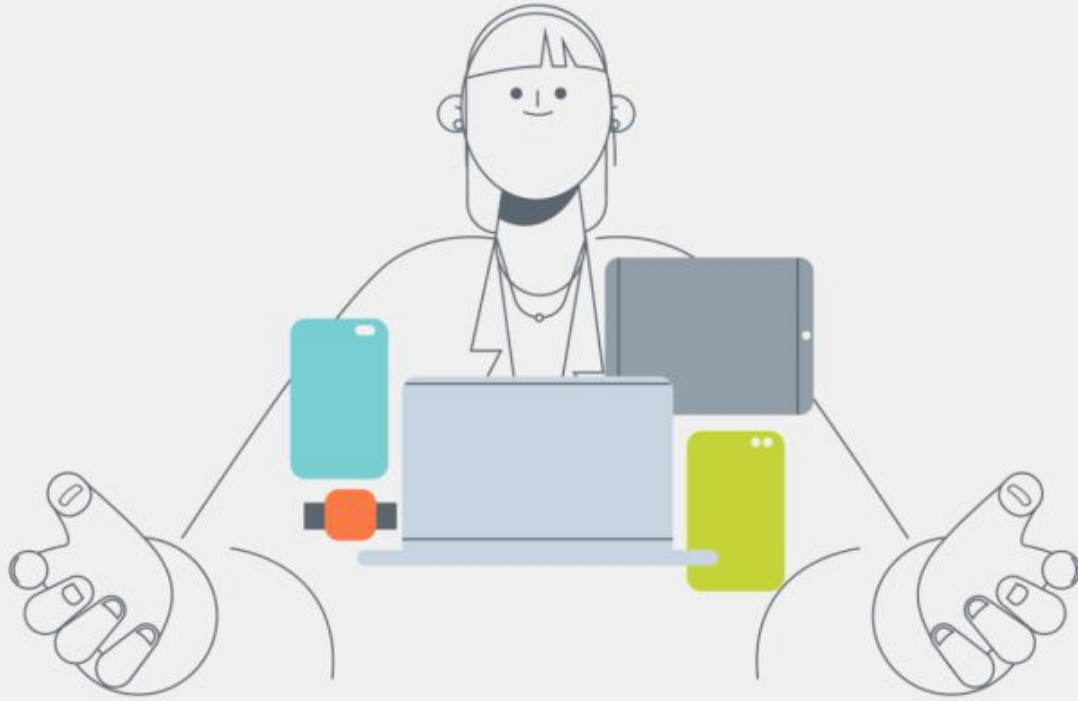
ultra low latency



It will have greater bandwidth, giving higher download speeds, eventually up to 10 gigabits per second (Gbit/s).



It will compete with existing ISPs, and also will make new applications possible in internet of things (IoT) and machine to machine areas.



4G cellphones are not able to use the new networks,
which require 5G enabled wireless devices.



The increased speed is achieved partly by using additional higher-frequency radio waves in addition to the low and medium band frequencies used in previous cellular networks.

The 5G spectrum is a range of radio frequencies in the sub-6 GHz range and the millimeter-wave (mmWave) frequency range that is 24.25 GHz and above.

The 5G spectrum refers to the radio frequencies that carry data from user equipment (UE) to cellular base stations to the data's endpoint.

5G NR

5G NR is the latest generation of mobile networks, providing ultra-fast speeds, low latency, and support for massive IoT (Internet of Things) applications.

Examples:

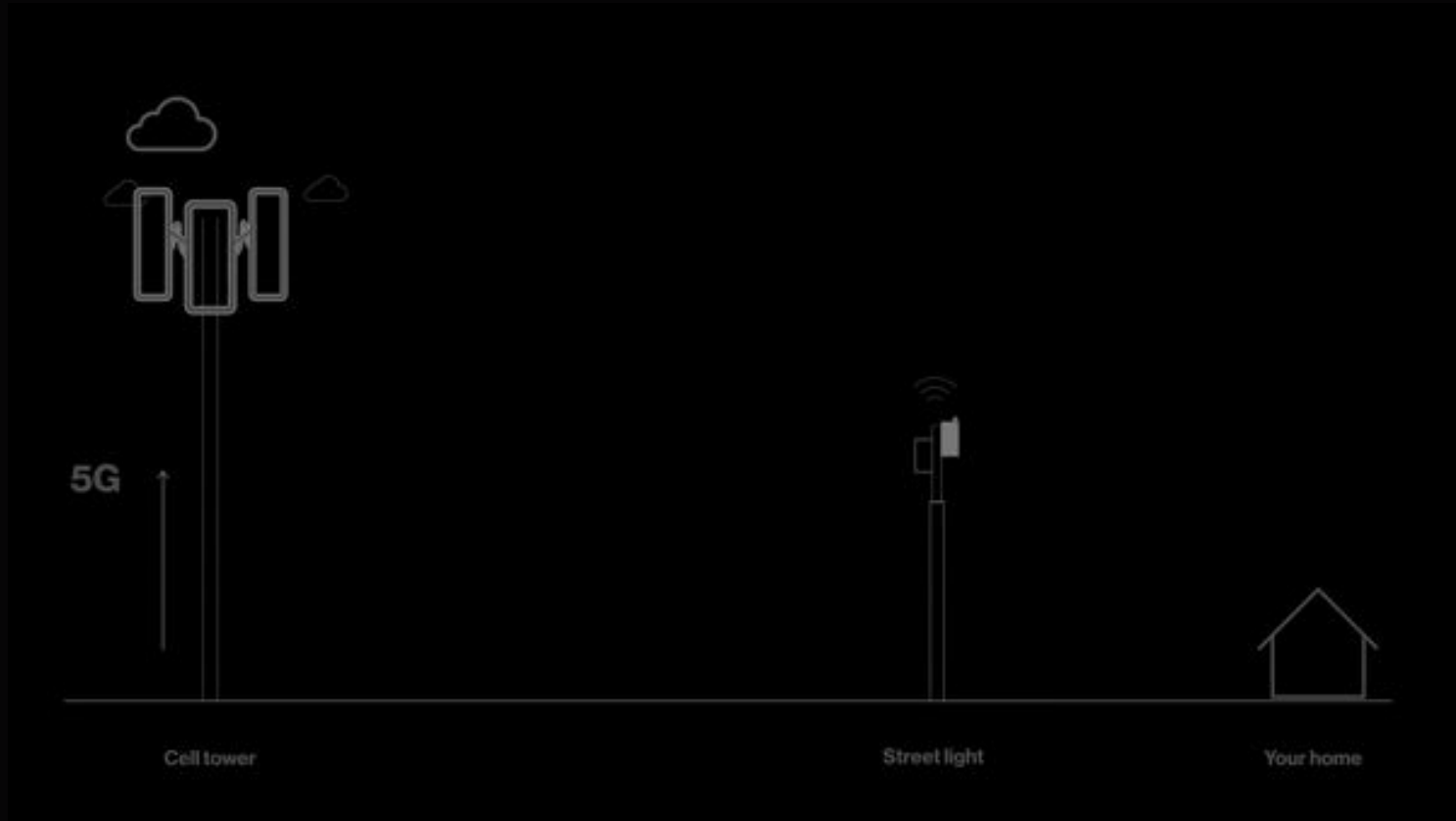
- 5G mobile networks for smartphones
- Smart city infrastructure (connected traffic lights, sensors)
- Augmented Reality (AR) and Virtual Reality (VR) applications



4G

Higher-frequency radio waves have a shorter useful physical range, requiring smaller geographic cells.

5G



A 5G network will be composed of networks consisting of up to three different types of cells, each requiring specific antenna designs as well as providing a different tradeoff of download speed to distance and service area.





For wide service,
5G networks operate on up to three frequency bands



5G

6 GHz

24 GHz

100 GHz

4G

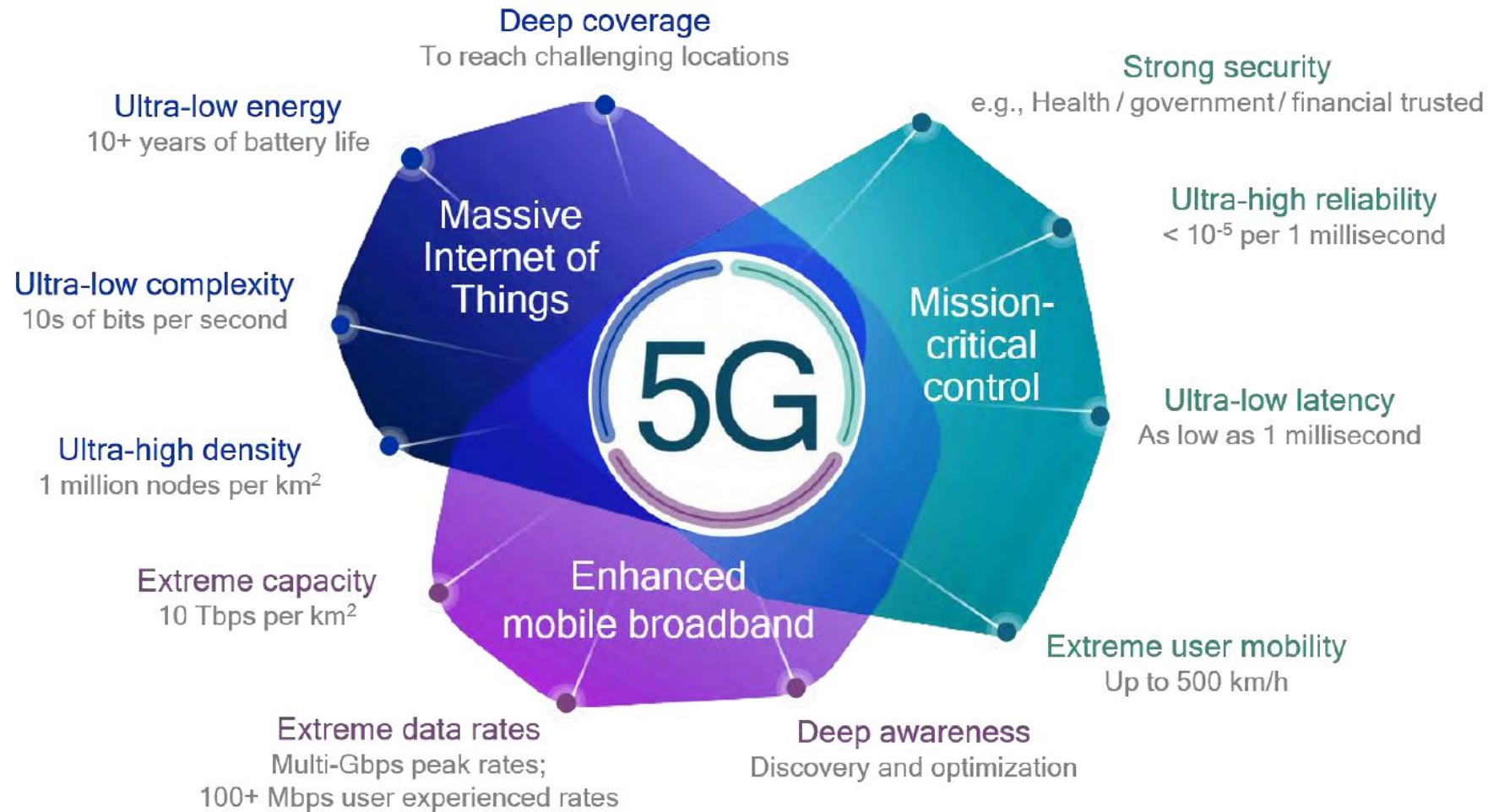
Large Millimeter Wave Bandwidth Opportunity



Application areas

— — —

- Enhanced Mobile Broadband (eMBB)
- Ultra Reliable Low Latency Communications (URLLC)
- Massive Machine Type Communications (mMTC)





Advantages & Disadvantages

— — —

Advantages:

Ultra-high speeds (up to 10 Gbps)

Extremely low latency (<1ms)

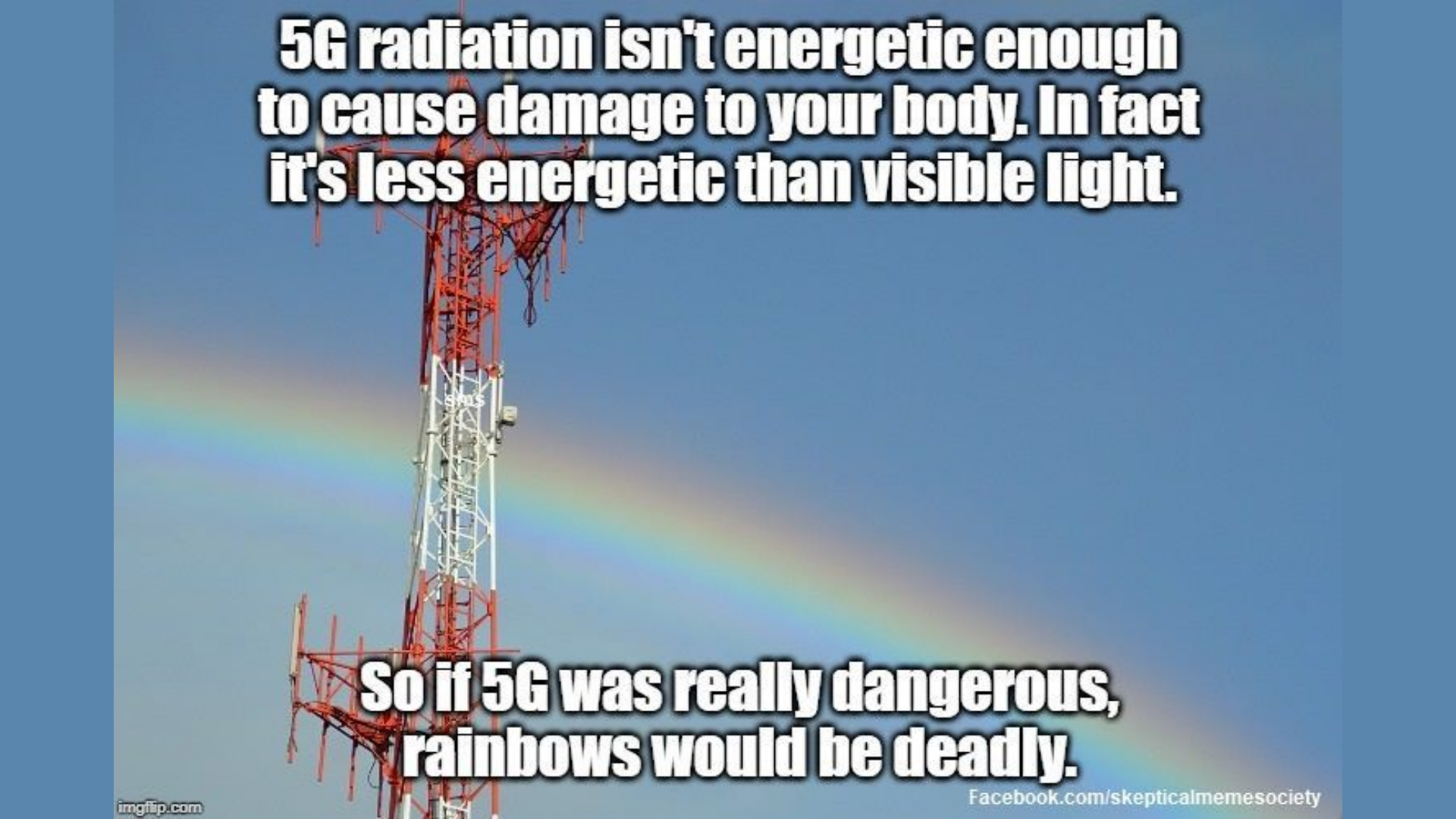
Supports a large number of connected devices

Disadvantages:

Requires new infrastructure (expensive deployment)

Limited coverage in many areas

Potential security and privacy risks

A tall, red and white telecommunications tower stands against a clear blue sky. A faint, multi-colored rainbow is visible in the background, arching across the sky. The tower has various antennas and equipment attached to its structure.

**5G radiation isn't energetic enough
to cause damage to your body. In fact
it's less energetic than visible light.**

**So if 5G was really dangerous,
rainbows would be deadly.**

THE EVOLUTION OF MOBILE TECHNOLOGY

2007



**USES 2G NETWORK
TO SHARE SCARY
ARTICLE ABOUT
DANGERS OF 3G**

2010



**USES 3G NETWORK
TO SHARE SCARY
VIDEO ABOUT
DANGERS OF 4G**

2018



**USES 4G NETWORK
TO SHARE SCARY
HD VIDEO ABOUT
DANGERS OF 5G**

WWW Wireless World Wide Web



Wireless World Wide Web

The Wireless World Wide Web (WWW) refers to the ability to access the internet wirelessly, without the need for physical cables or wired connections. It is powered by technologies such as Wi-Fi, cellular networks (3G, 4G, 5G), and satellite internet.

Key Technologies Behind the WWW

Wi-Fi – Enables wireless internet access through local area networks (LANs).

Cellular Networks (3G, 4G, 5G, and Beyond) – Provides mobile broadband access.

Satellite Internet – Delivers connectivity to remote areas.

Bluetooth – Supports short-range wireless communication.

Li-Fi (Light Fidelity) – Uses light waves for ultra-fast data transmission.



Security Issues related to Wireless Communications

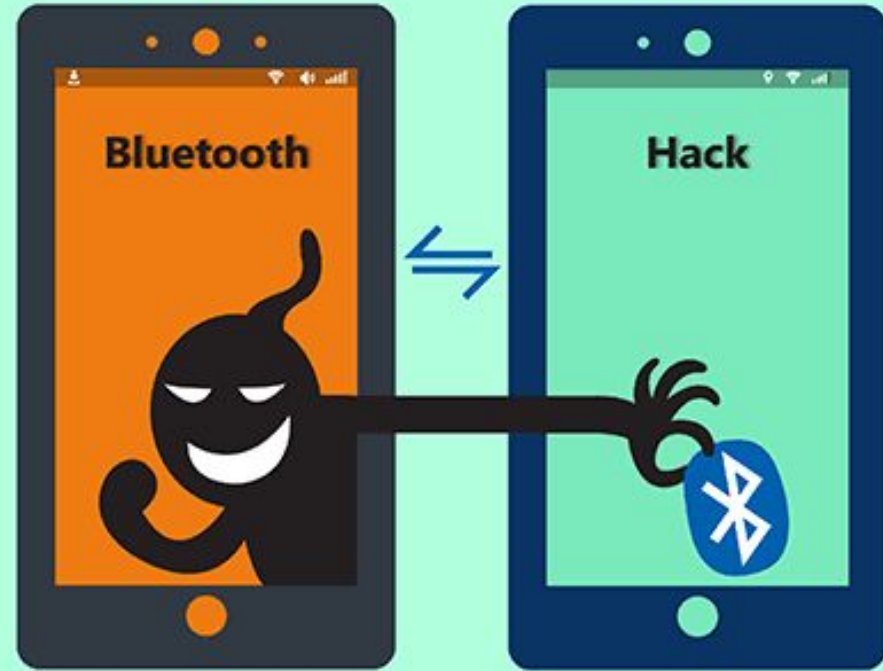
Bluetooth Attacks

- Bluejacking

Unsolicited messages

- Bluesnarfing

Stealing information



Bluejacking

This is like receiving a prank message from a stranger via Bluetooth. The attacker sends unwanted messages to nearby devices, but they can't steal your data.

Bluesnarfing

This is more dangerous.
Hackers secretly connect
to your device and steal
personal information
like contacts, messages,
and files without you
knowing.

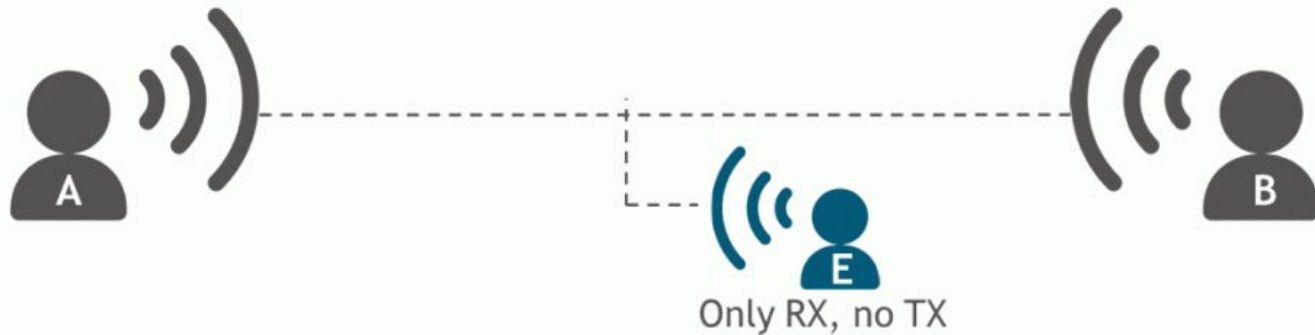
NFC attacks

Contactless credit cards or other NFC impersonation / attacks.

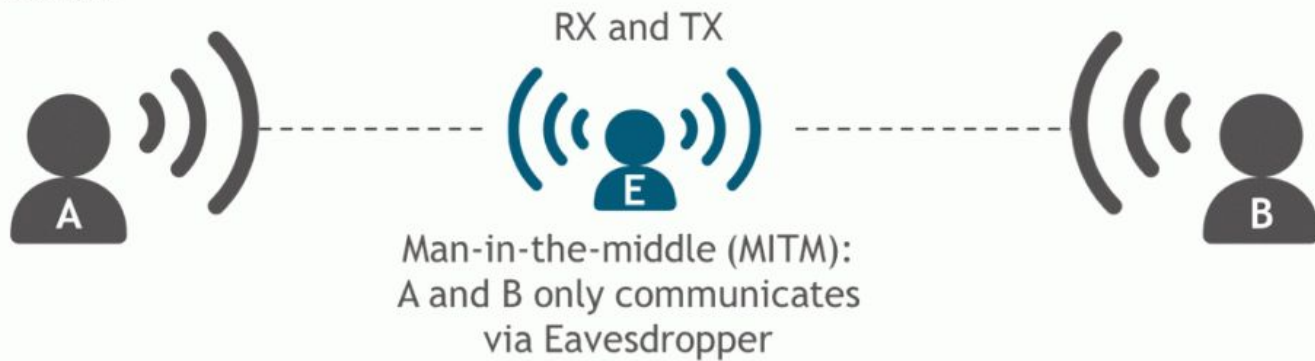


Eavesdropping

Passive



Active



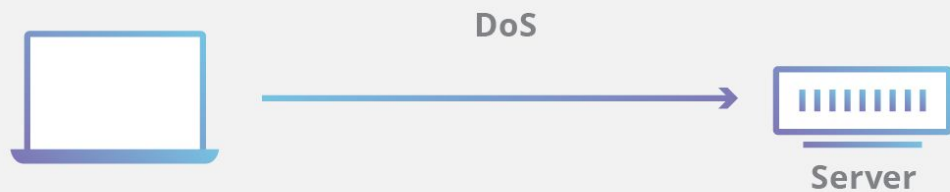
Eavesdropping

Eavesdropping is the act of secretly listening to the private conversations of others without their consent.

It can be done in person (e.g., standing near a door or window) or electronically (e.g., intercepting phone calls, hacking into communication channels).

Denial of Service attacks

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Denial of Service attacks

A Denial of Service (DoS) attack is a cyberattack aimed at making a system, network, or website unavailable to users by overwhelming it with excessive traffic or requests.

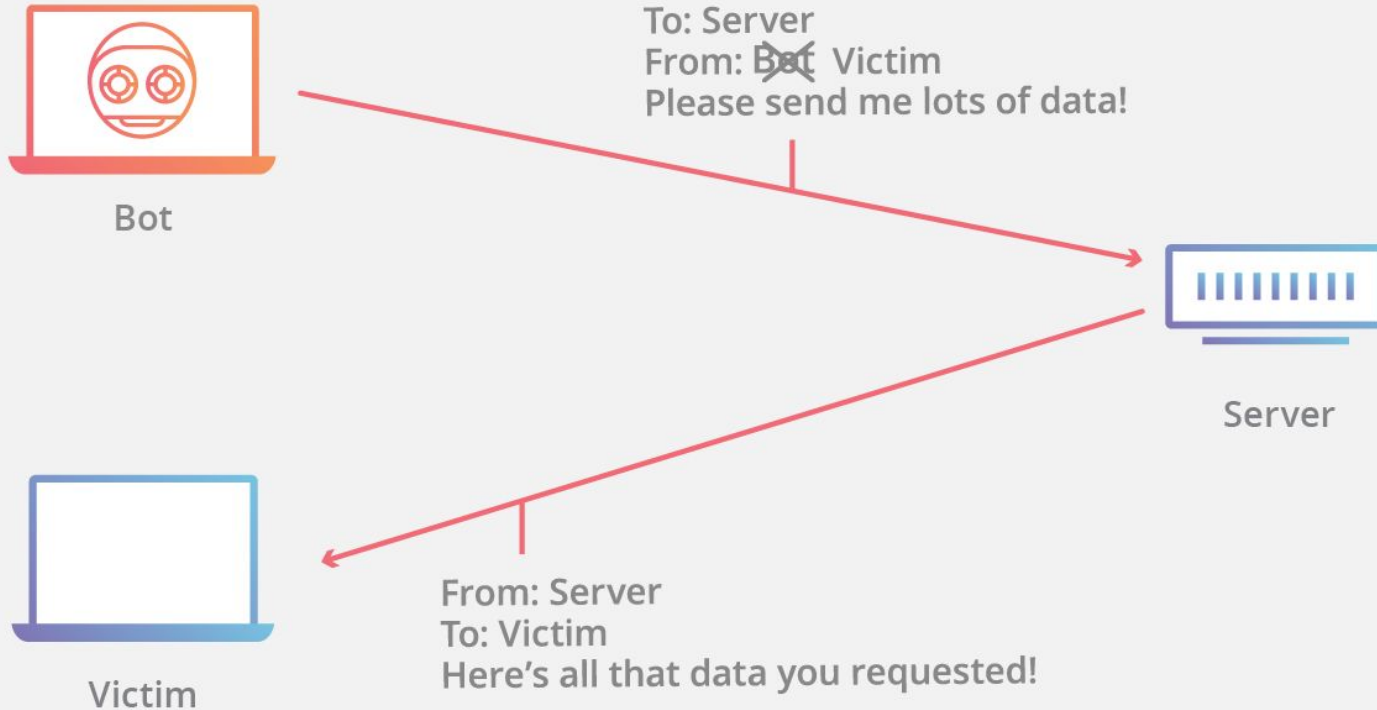
when I attack my friend with ddos



BRACE YOURSELVES

**THE PACKETS ARE
COMING**

Spoofing and Session Hijacking



Spoofing

— — —

Spoofing is when a hacker pretends to be someone or something else to trick people or systems.

IP Spoofing – A hacker fakes their computer's address to hide their identity or attack a network.

Email Spoofing – You get an email that looks like it's from a friend or company, but it's actually from a hacker trying to scam you.

Caller ID Spoofing – A phone call appears to be from a trusted number, but it's actually a scammer.

Website Spoofing – A fake website looks exactly like a real one to steal your login details (e.g., a fake bank website).

Session hijacking

When you log in to a website, the site gives you a session ID (a special key) to keep you logged in. Session hijacking happens when a hacker steals this key and takes over your session.

Man-in-the-Middle Attack - A hacker secretly listens to your internet traffic and steals your session ID.

Stealing Cookies - Websites save small files (cookies) to remember you. If hackers steal these, they can log in as you.

Fake Login Pages - You enter your details on a fake page, and hackers steal them.

Victim



Login: user1, password: 1234



Session ID: F7299EBA209CC23

Server



Attacker

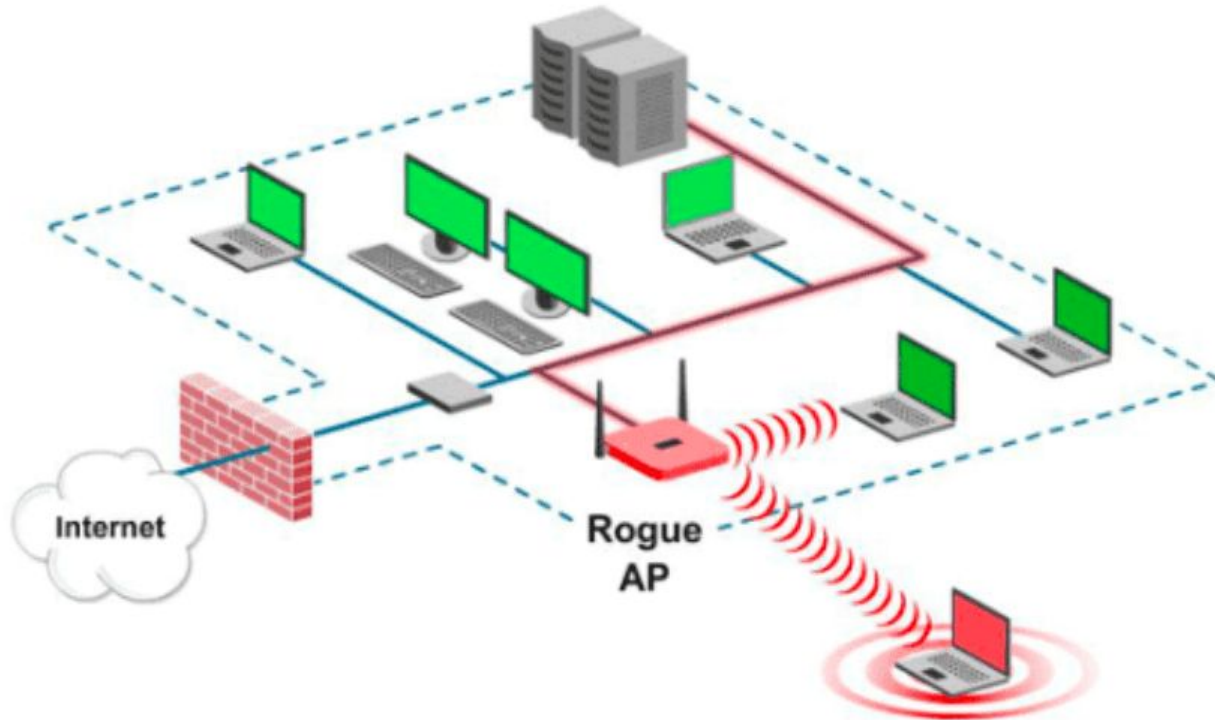
Session ID: F7299EBA209CC23



LOOK AT ME

I AM THE CAPTAIN NOW

Rogue access point



Rogue access point

A Rogue Access Point is a fake Wifi network set up by a hacker to trick people into connecting to it. Once connected, the hacker can steal personal information, track online activities, or inject malware into devices.

1. A hacker sets up a Wi-Fi network with a familiar name (e.g., "Free Coffee Wifi" or "Library Wifi").
2. People connect to it, thinking it's a safe network.
3. The hacker can now see what they are doing online and steal login details, credit card information, or other sensitive data.

Passive Capturing

Passive capturing is when a hacker silently listens to network traffic to steal information. Unlike active attacks, the hacker doesn't interfere with the network—just secretly collects data.

For **example**, if you're at a coffee shop using public Wi-Fi, a hacker nearby can use a tool like Wireshark to passively capture your unencrypted data, such as login credentials or personal messages, as they travel over the network. The hacker doesn't interfere directly but simply listens in on the traffic.

Understanding WEP

- WEP relies on a secret key which is shared between the sender (mobile station) and the receiver (access point).
- Secret Key : packets are encrypted using the secret key before they are transmitted.
- Integrity Check : it is used to ensure that packets are not modified in transit

WEP (Wired Equivalent Privacy)

Encryption:

40 / 64 bits

104 / 128 bits

24 bits are used for IV (Initialization vector)

Passphrase:

Key 1-4 | Each WEP key can consist of the letters "A" through "F" and the numbers "0" through "9". It should be 10 hex or 5 ASCII characters in length for 40/64-bit encryption and 26 hex or 13 ASCII characters in length for 104/128-bit encryption.

HELL YEAH!



**FOUND A WEP ENCRYPTED
WIFI NETWORK**

MAC filtering

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Hidden SSIDs

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Wi-Fi protected access (WPA/WPA2)

Wi-Fi Protected Access (WPA), Wi-Fi Protected Access II (WPA2), and Wi-Fi Protected Access III (WPA3) are the three security and security certification programs developed by the Wi-Fi Alliance to secure wireless computer networks.

The Alliance defined these in response to serious weaknesses researchers had found in the previous system, Wired Equivalent Privacy (WEP).

Target users (authentication key distribution)

WPA-Personal

WPA-PSK (pre-shared key) mode, this is designed for home and small office networks and doesn't require server.

WPA-Enterprise

WPA-802.1X mode, and sometimes just WPA , this is designed for enterprise networks and requires a RADIUS server.

Wi-Fi Protected Setup

alternative authentication key distribution method intended to simplify and strengthen the process

Wi-Fi Protected Setup (WPS)



Four modes in a home network for adding a new device:

- PIN method
- Push button method
- Near-field communication method
- USB method

WPS is widely understood to be insecure.

It should not be used.

What's the WiFi password?

IMAGE

Check the back of the router

Back of the router:



WPA/WPA2 Personal

Encryption:

TKIP / AES

Pre-Shared Key:

A key of 8-63 characters

Key Renewal:

You can choose a Key Renewal period, which instructs the device how often it should change encryption keys.

The default is 3600 seconds.

WPA3

- Announced in 2018
- Uses an equivalent 192-bit cryptographic strength in WPA3-Enterprise mode
- Replaces the pre-shared key (PSK) exchange with Simultaneous Authentication of Equals (SAE) exchange
- WPA3 is required since July 1, 2020. The newer versions may not work with some older network cards.

Common defense techniques

- Change router default username and password
- Change the internal IP subnet if possible
- Change default name and hide broadcasting of the SSID (Service Set Identifier)
- None of the attack methods are faster or effective when a larger passphrase is used.
- Restrict access to your wireless network by filtering access based on the MAC (Media Access Code) addresses
- Use Encryption

IF YOU DO NOT REVISE...

YOU SHALL NOT PASS

Board Questions / Assignments

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1. I-Way, Components of I-Way
2. Internet, Extranet, Intranet (Differentiate)
3. Wireless communication, features of wifi that made it so popular in present
4. UMTS & LTE (Difference)
5. Role of I-Way
6. Working of WAP, Architecture of WAP
7. Wireless security issues
8. Explain all the wireless technologies